Cool Flames in Space



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Introduction

Cool flame: *A flame having a typical temperature of* ***~ 650 K****, reaction is not vigorous and releases* ***little heat, light, or carbon dioxide*** *(source: Wikipedia)*

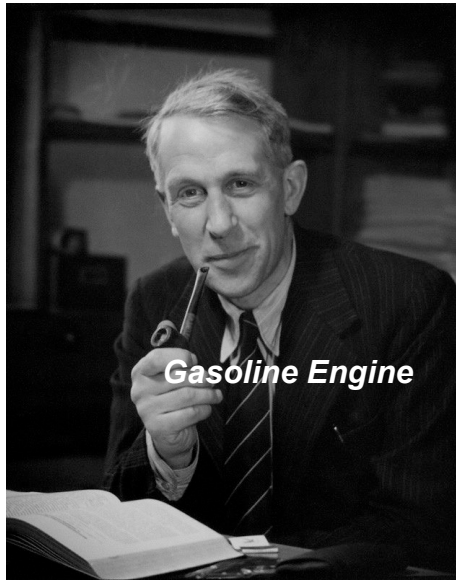


Sir Humphrey Davy

1817 Low temperature flame

Hot platinum wire into a mixture of air and diethyl ether vapor

Formation of pale phosphorescent light in the dark



Harry Julius Emeléus

First to record their emission spectra

1929 coined the term Cool /Cold flame

Introduction

Traditional view of cool flames prior to 2012-2014

- **Cool flames associated with premixed combustion leading to ignition of hot fuel/air mixtures – commonly encountered in car-engine knock**

Cool flames → Ignition

History of cool flame research – Professor Griffiths (2004)

Pre-1900 *First observation: Davy (1817), Perkin (1882)*

Poincore theorems and Liapounov stability analysis

1920 *First flow tube experiments*

Lotka-Volterra isothermal oscillations

1930 *Pease C_3H_8 cool flame studies in a closed vessel*
Extensive mapping of p-T ignition diagrams (Townsend et al.)

1940

Zeldovich chemical reactor theory

1950 **Sal'nikov non-isothermal theory**

$RH + O_2$ product analysis and $R + O_2 = RO_2$ mechanisms proposed

1960

Gray and Yang unified chain-thermal theory

1970

Thermokinetic models

Stirred closed vessel and CSTR studies

Chemical reactor stability and non-linear dynamics (Aris, Uppal et al.)

1980

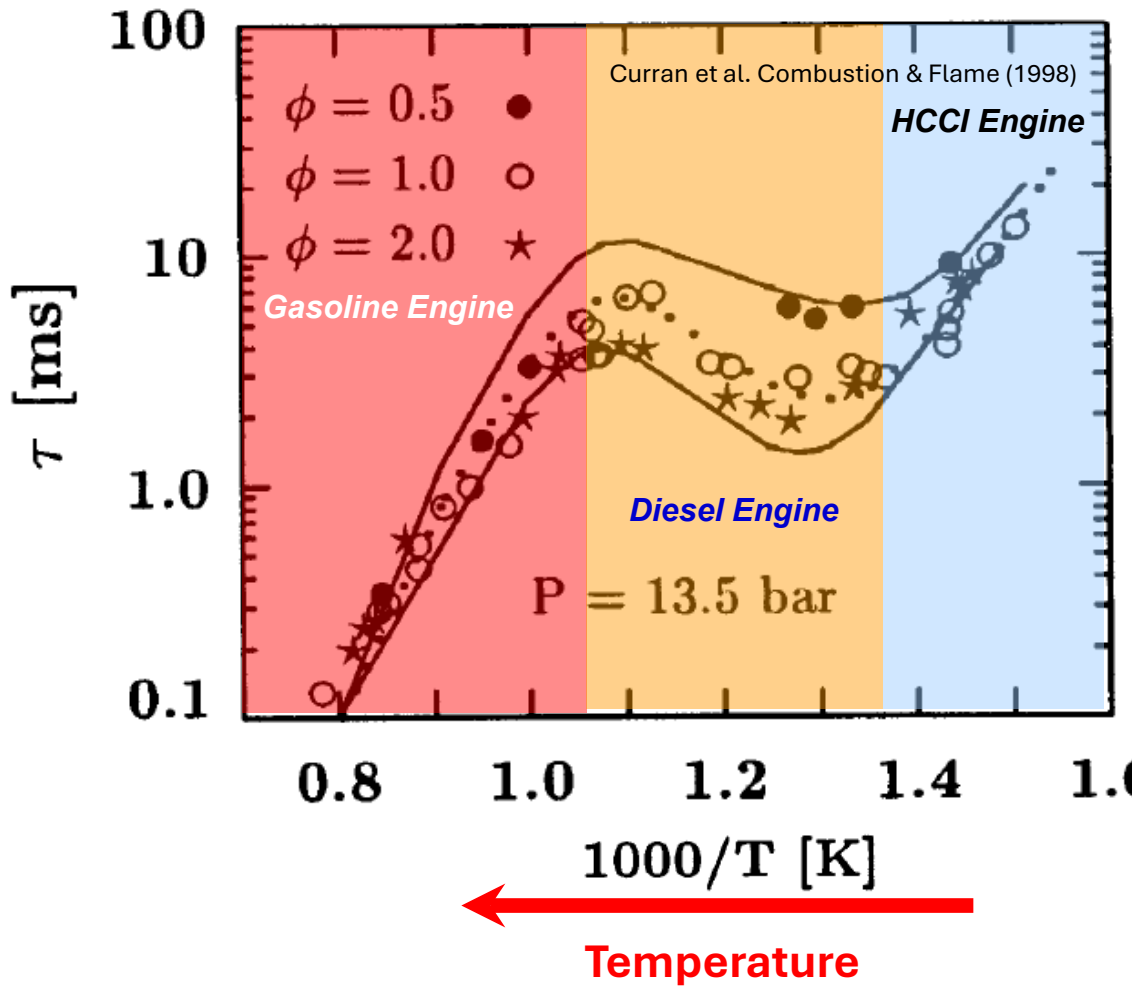
1990 *Zero dimensional numerical simulation and reduced/detailed kinetic model development*

Technical development and applications

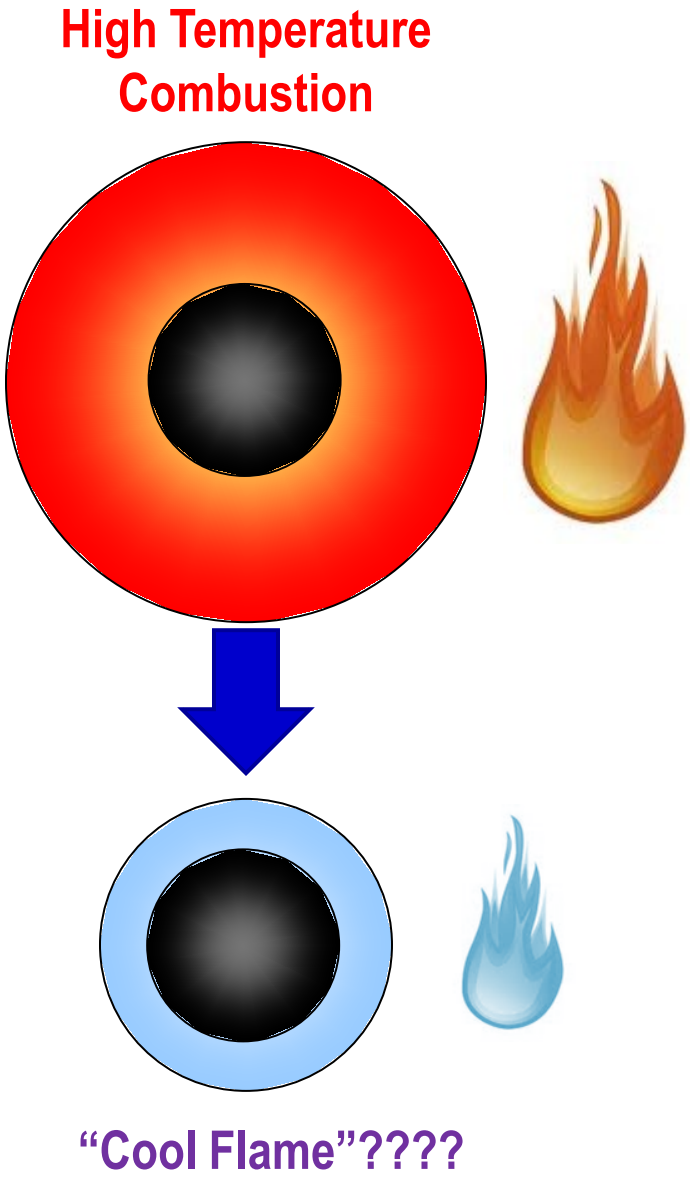
2000 *Reaction – diffusion models*

Cool flames in space (μg)

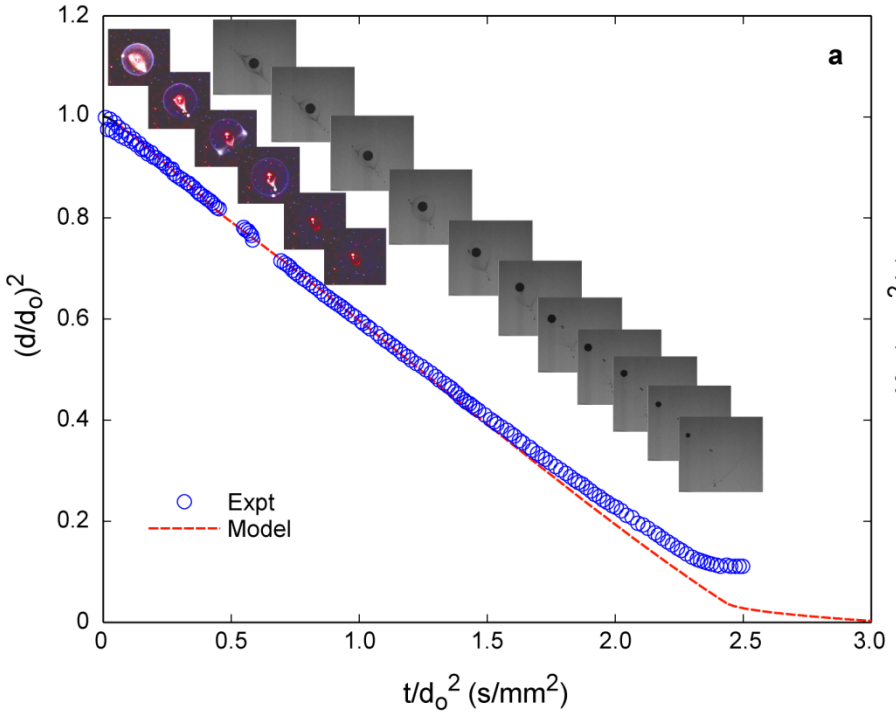
Ignition delay of n-heptane

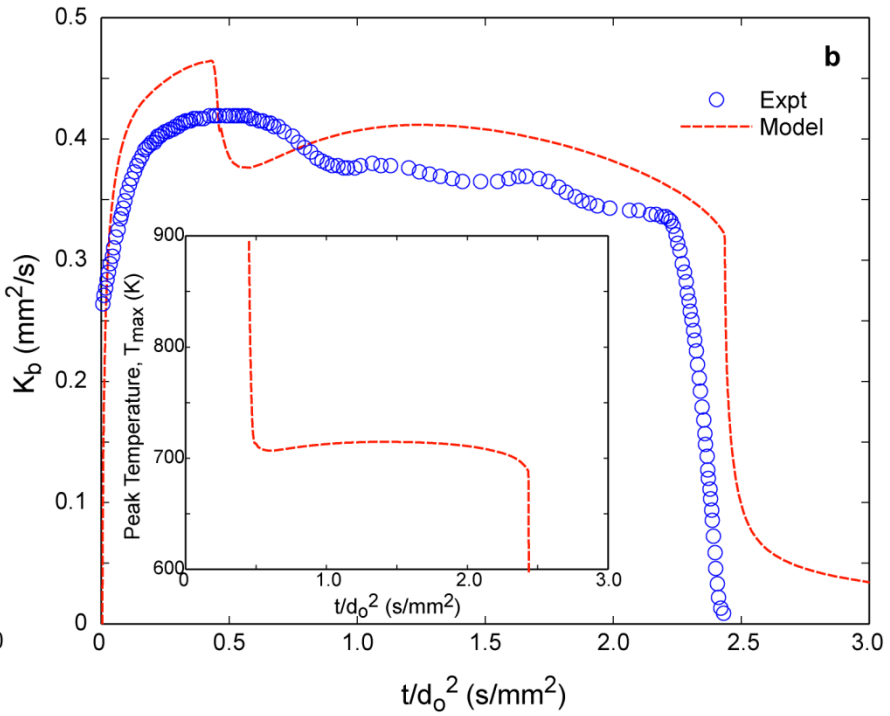


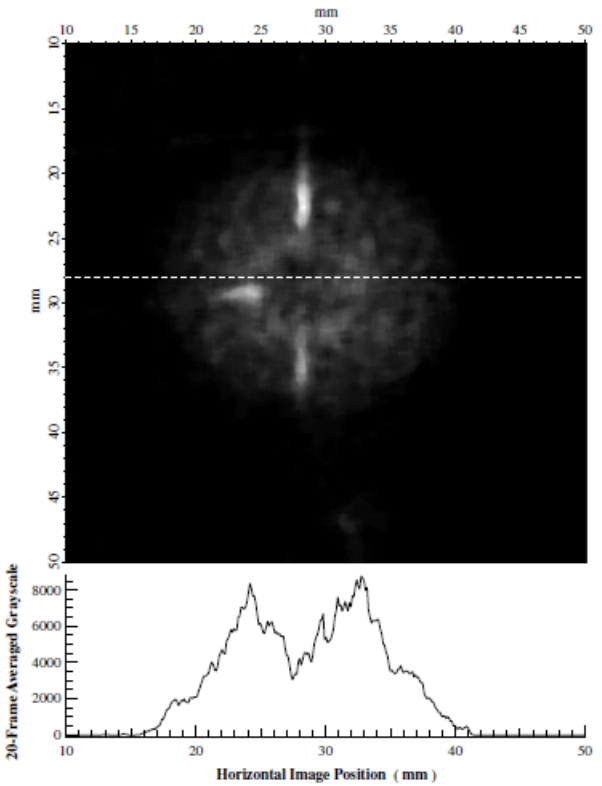
First observation of multiphase cool flames – “Anomalous Combustion”



Cool Flames – Model Predictions

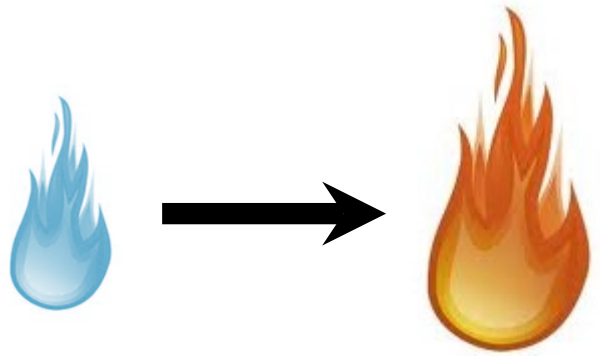




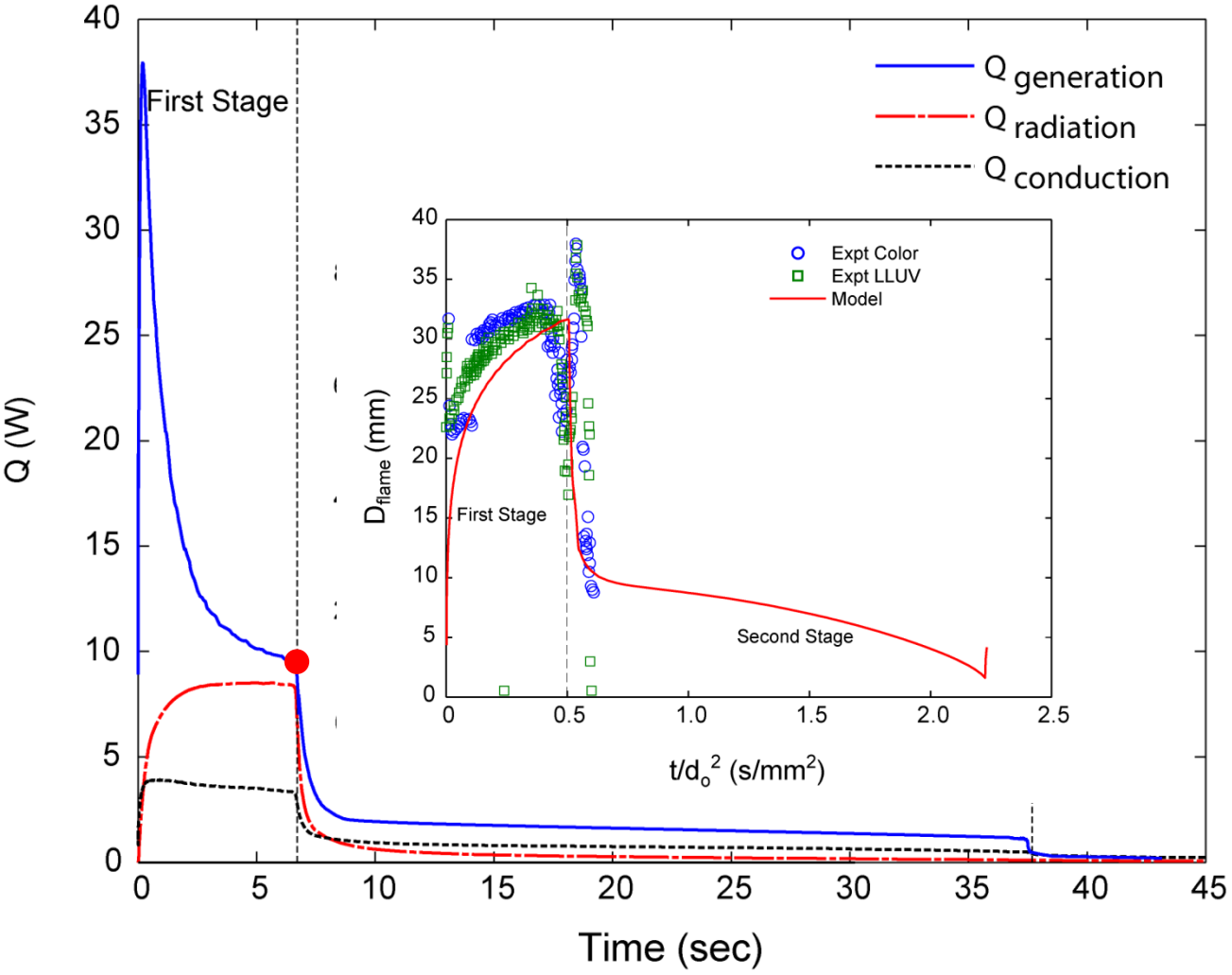


Evolution of droplet diameter and burning rate (n-heptane, d_o = 4.00 mm, air, atmospheric pressure)

Classical *"Cool Flames"* (premixed systems) —*"Cool Flame"* to Hot Ignition.



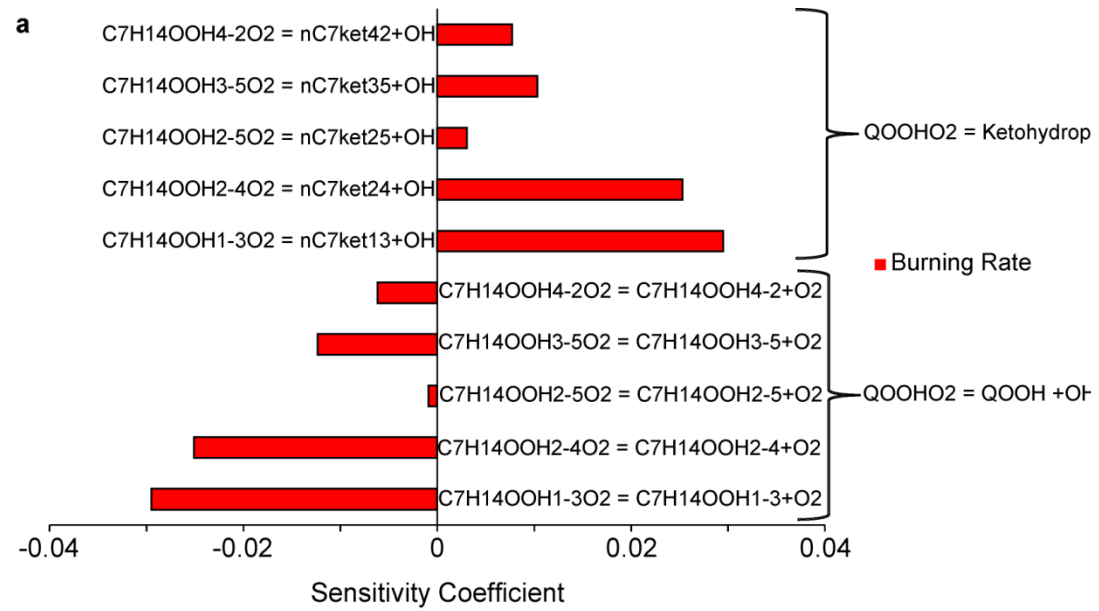
Driving Mechanism

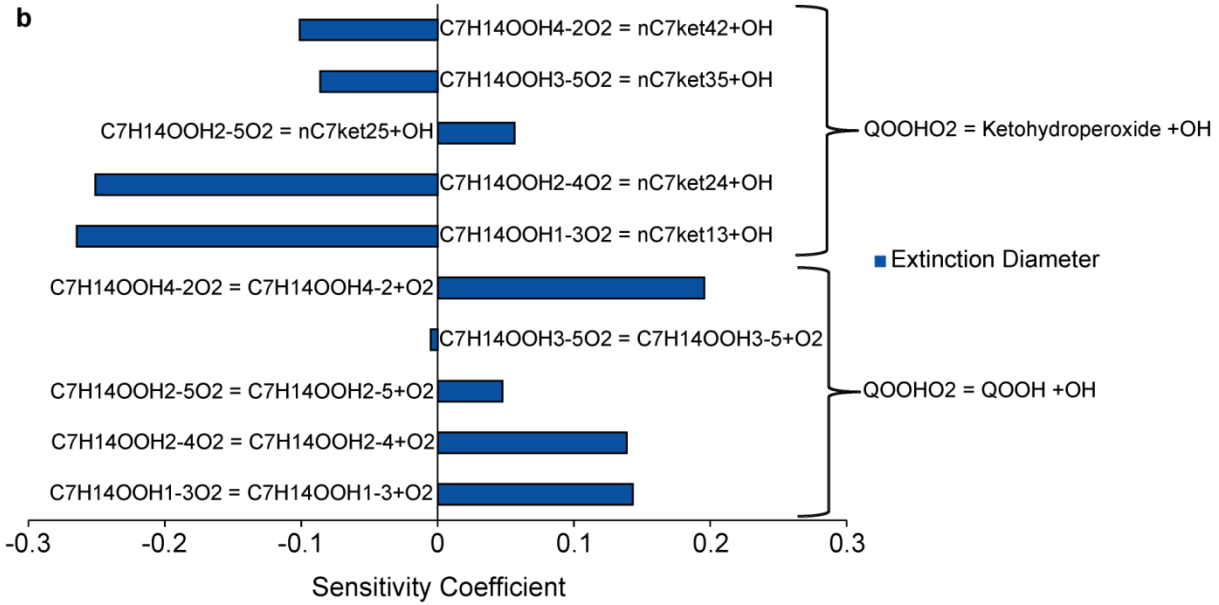


Evolution of heat generation and losses (n-heptane, d_o = 4.00 mm, air, atmospheric pressure)

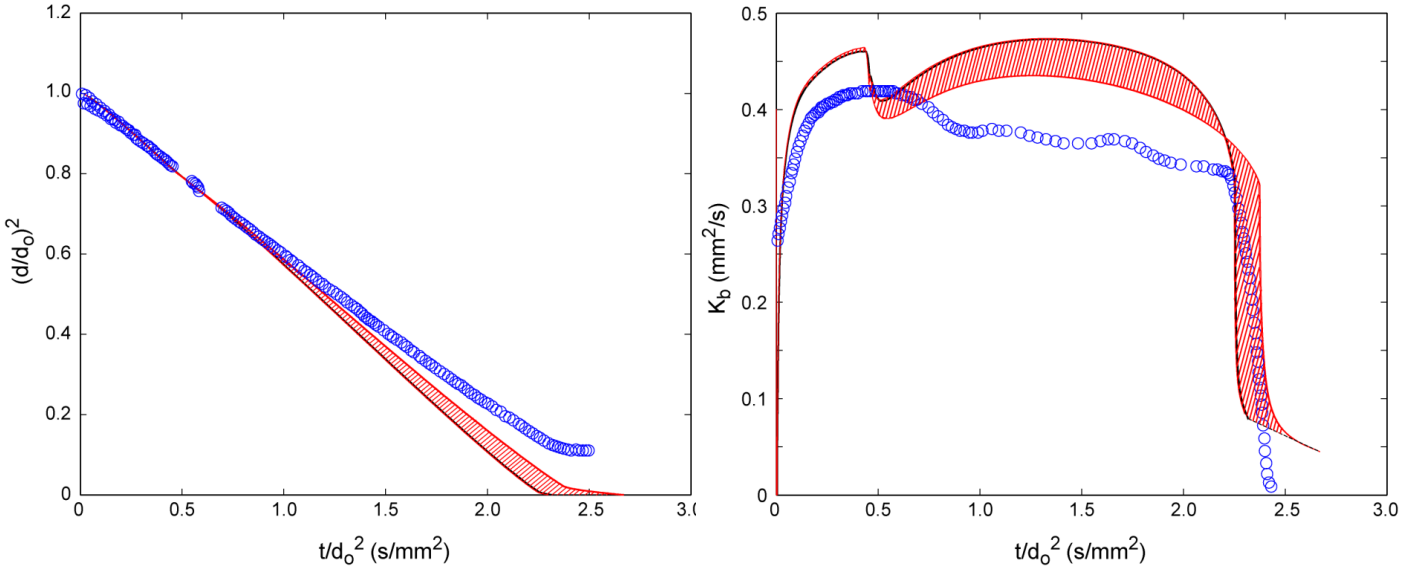
- Radiative heat loss pushes the system to transition to the second stage, cool flame burn.
- A dynamic balance between heat generation and heat loss maintains the long-duration cool flame burn.

Sensitivity of Kinetic Parameters





Sensitivity coefficients for burning rate and extinction diameter predictions.

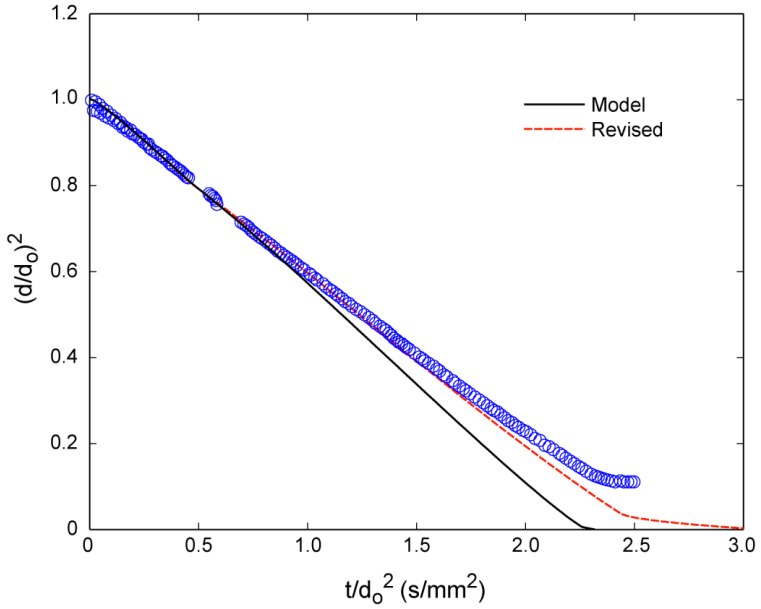


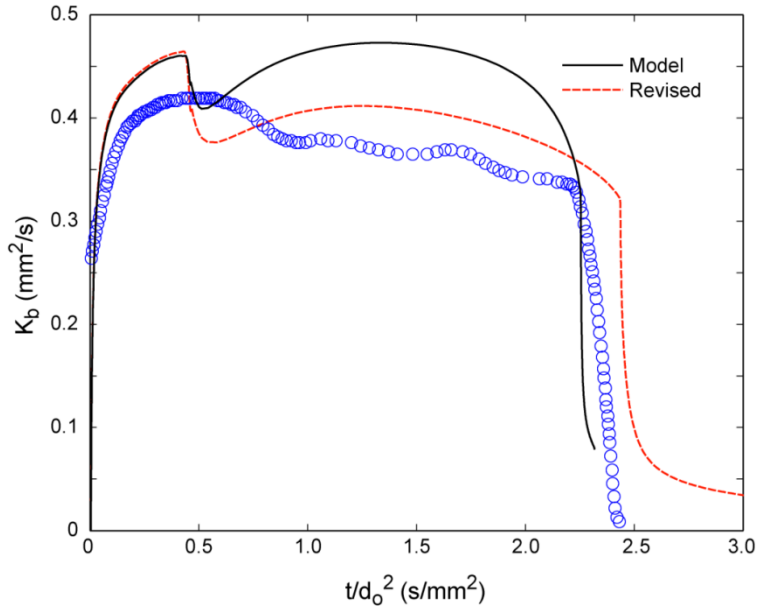
Uncertainty in droplet diameter and burning rate predictions, (n-heptane, d_o = 4.00 mm, air, atmospheric pressure)

In general increase in $QOOHO_2 \rightarrow QOOH + O_2$ rate decreased the burning rate and increased the extinction diameter.

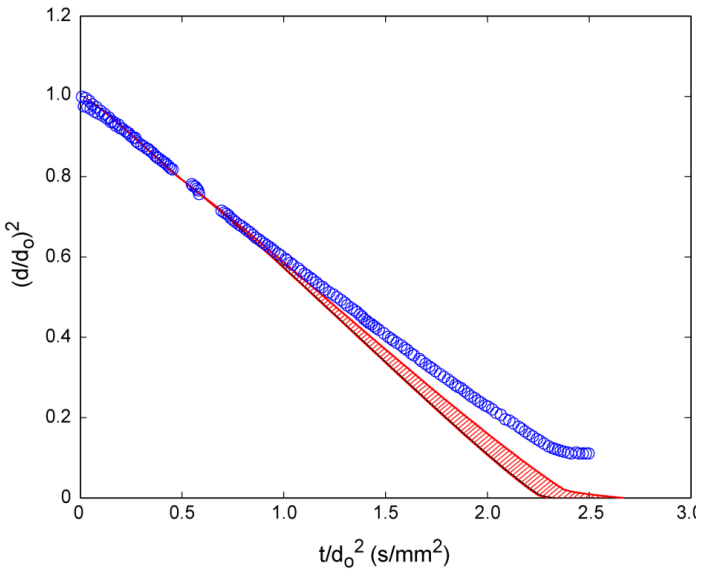
Vice versa $QOOHO_2 \rightarrow$ Ketohydroperoxide + OH ($Q = C_nH_{2n}$)

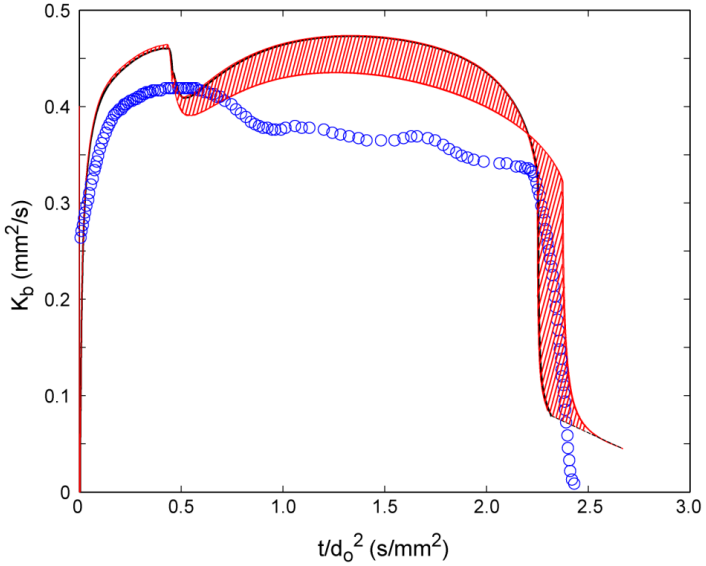
Cool Flames → Revised Kinetics





Predicted droplet diameter and burning rate evolution with revised chemical kinetics model.





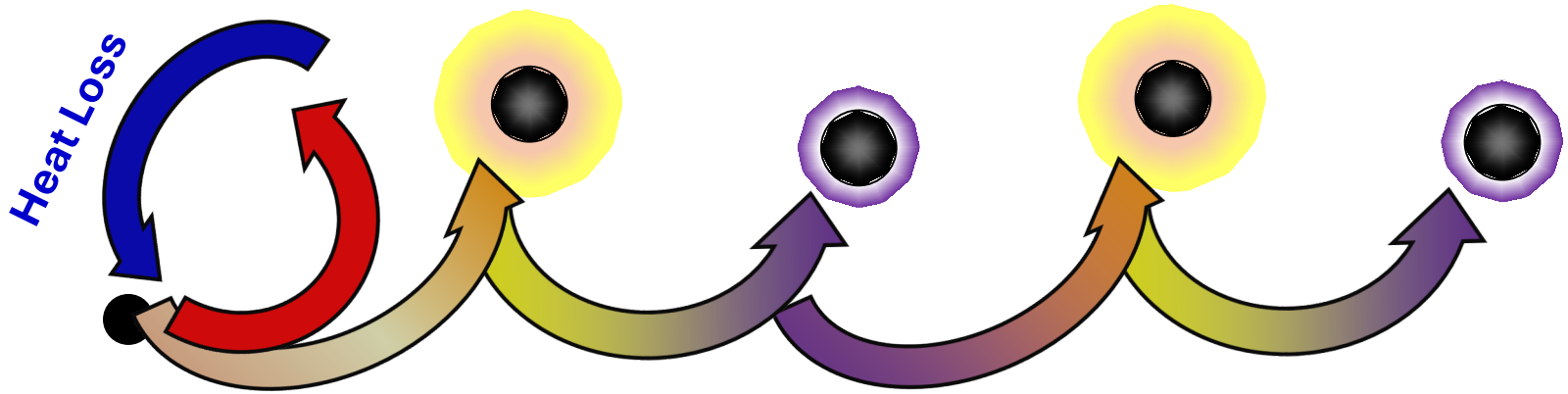
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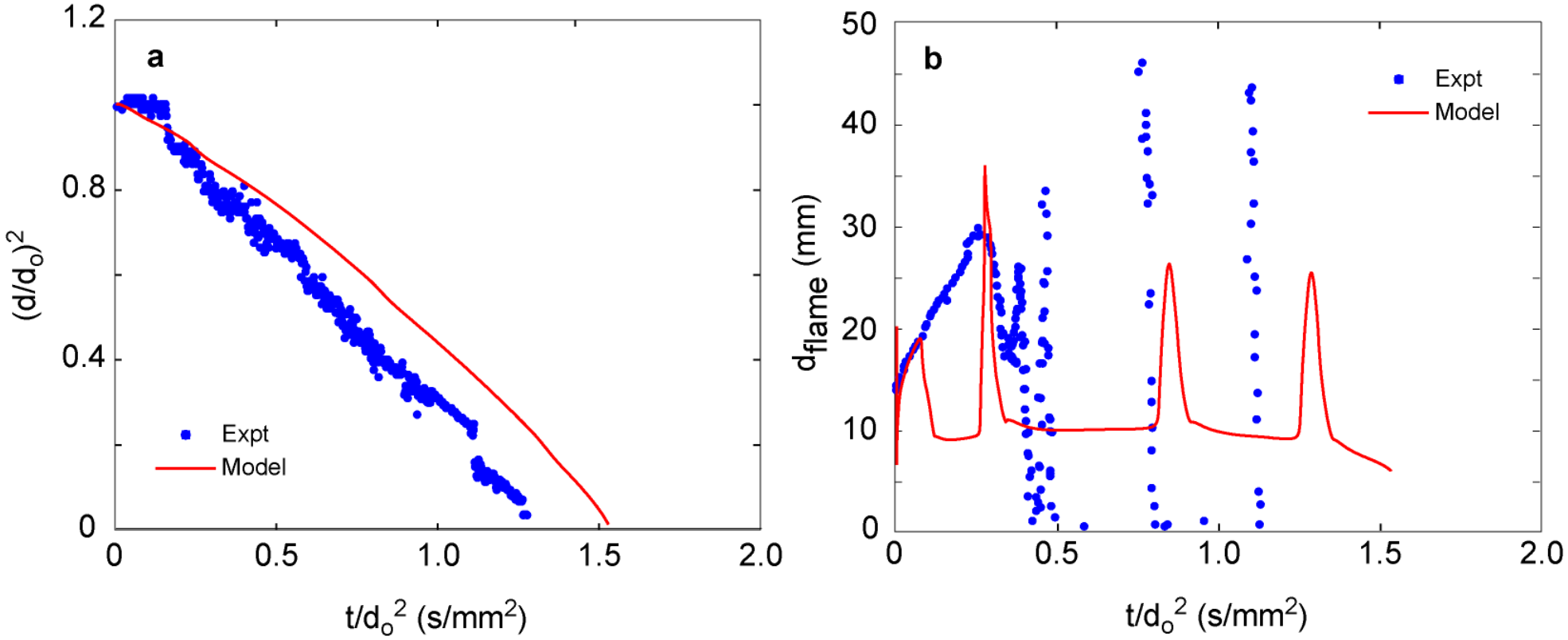
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Multistage Oscillatory Cool Flames – Pressure Effects

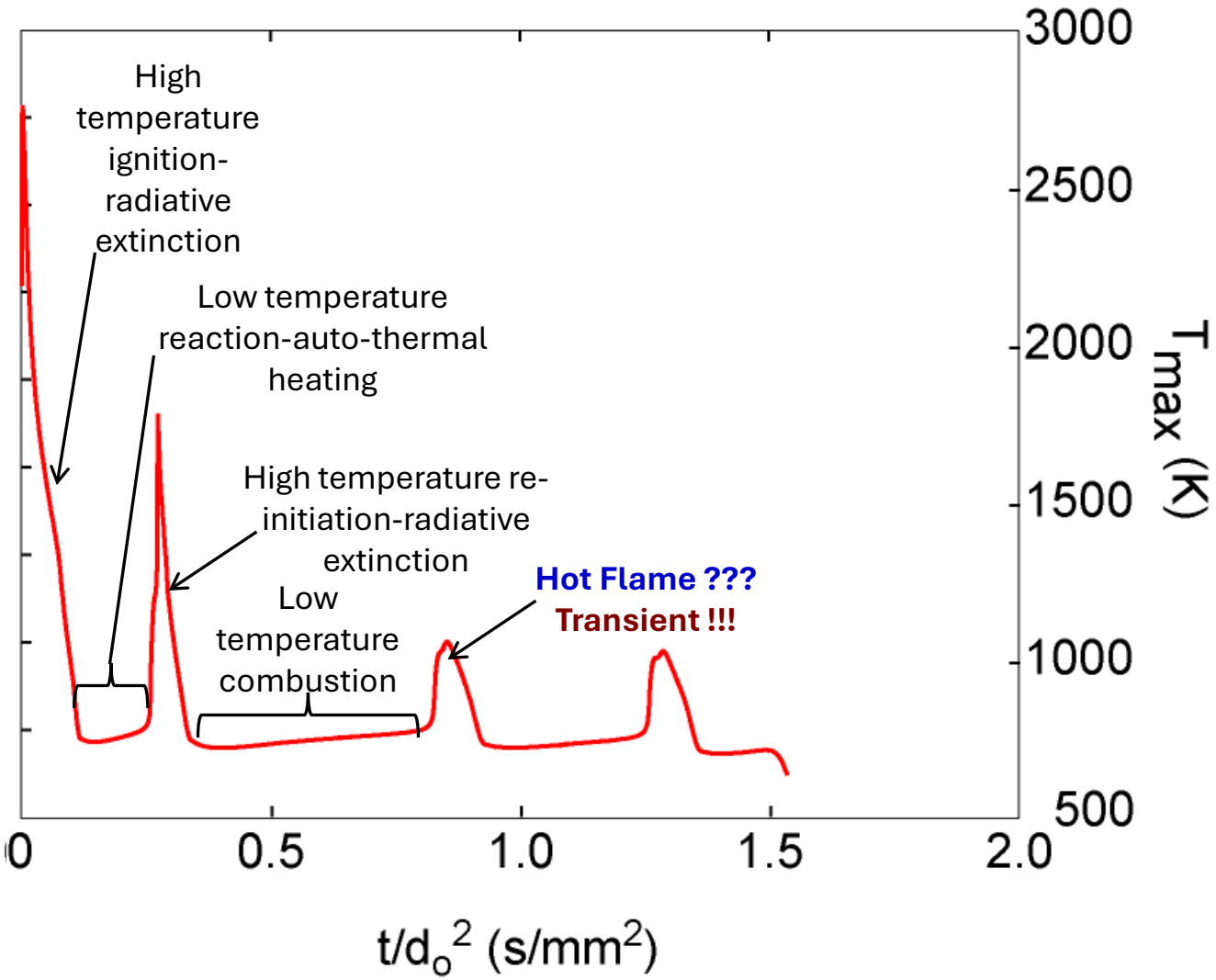
Oscillatory Cool Flames – Model Predictions



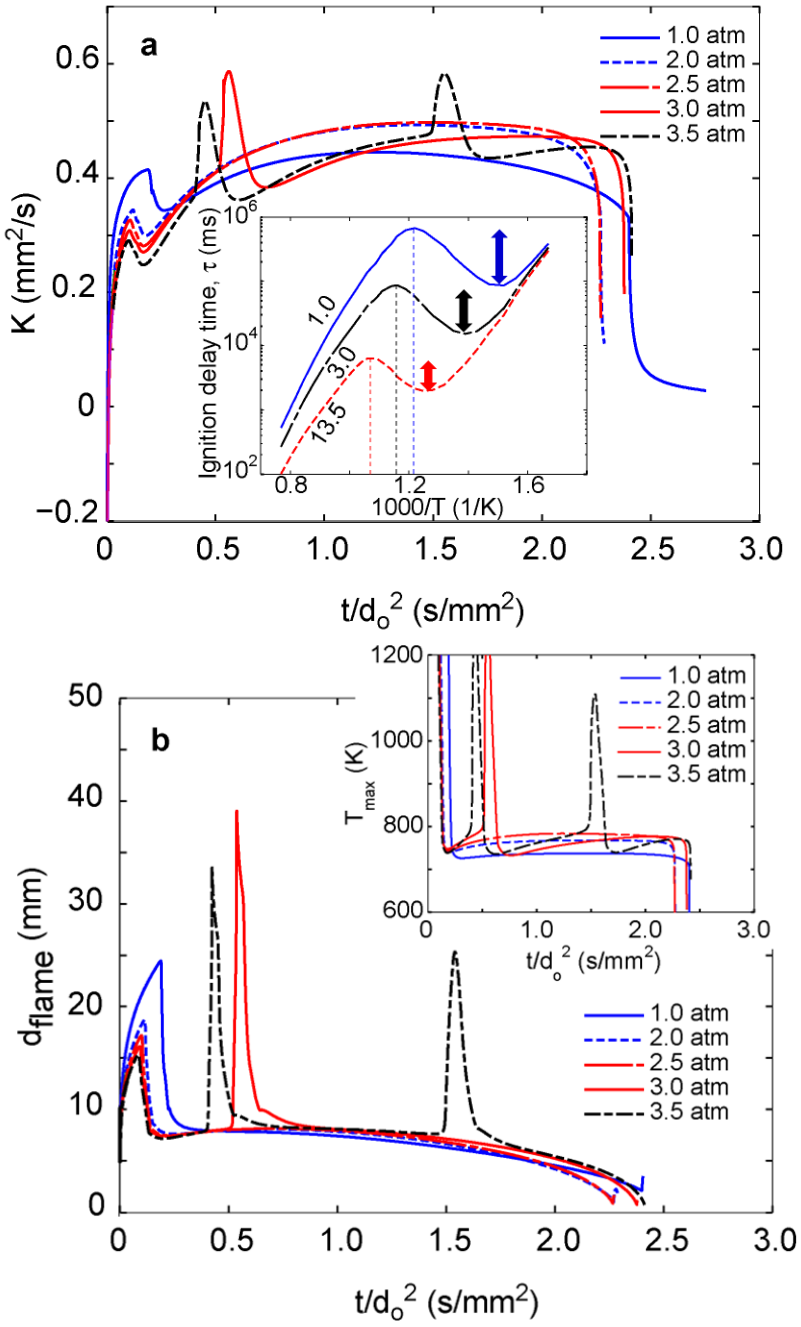


Evolution of droplet diameter and flame diameter (d_o = 4.08 mm, $O_2/CO_2/N_2$ 0.21/0.15/0.64, 3 atm)

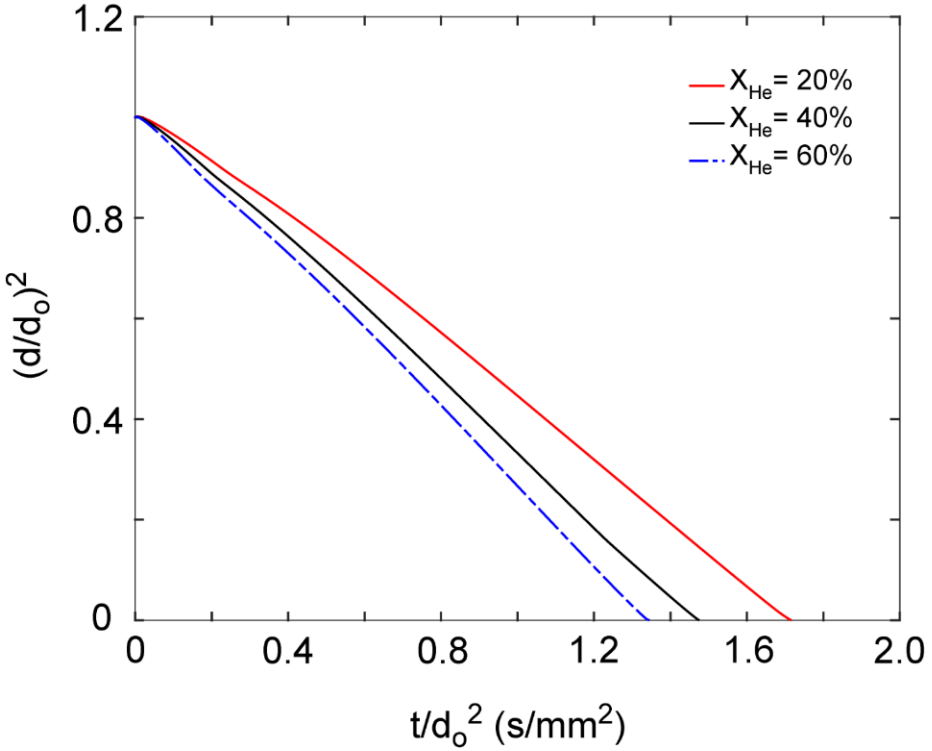
Oscillatory Cool Flames – Model Predictions



Evolution of maximum temperature and burning rate (d_o = 4.08 mm, $O_2/CO_2/N_2$ 0.21/0.15/0.64, 3 atm)

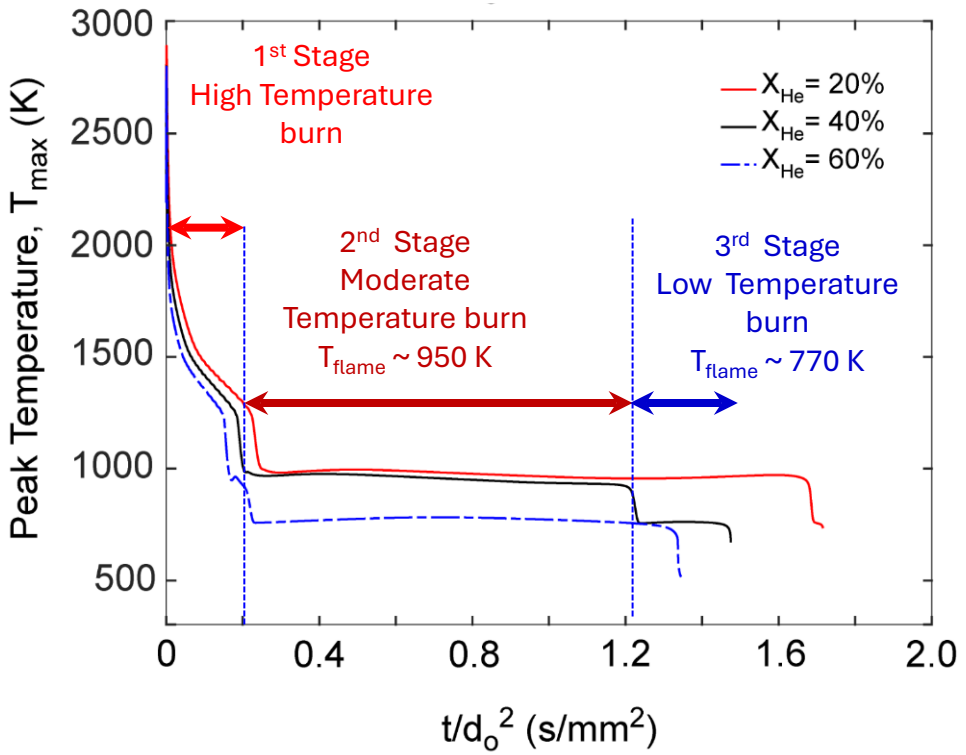


High Pressure Helium Exchange – 3 atm



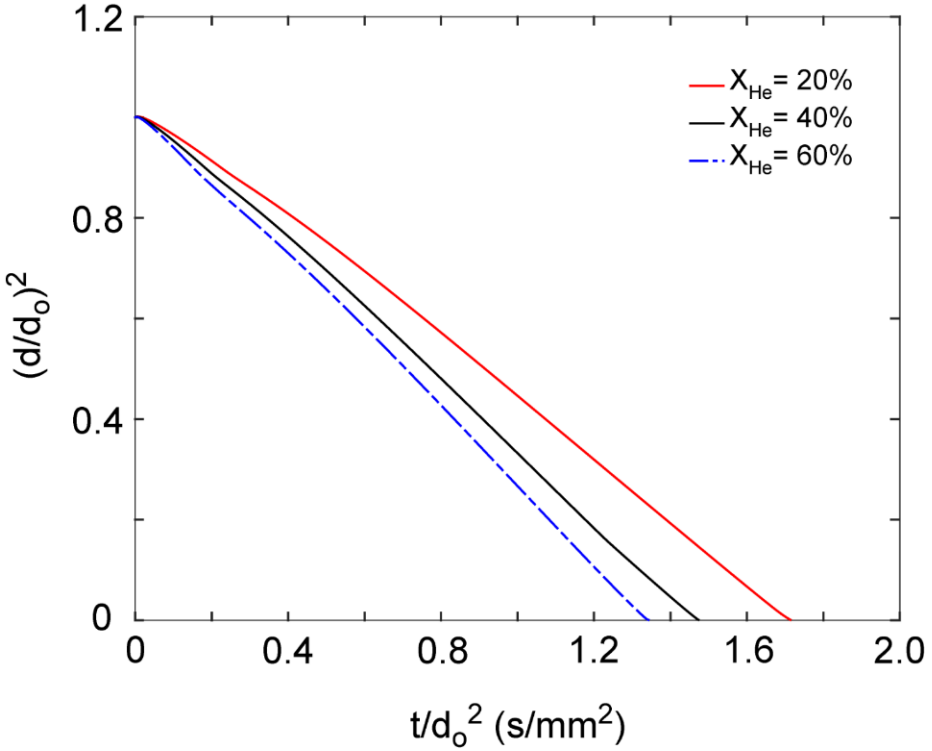
Evolution of droplet diameter (n-heptane, d_o = 4.00 mm, X_{O2} = 21%, He exchange with balance N_2, 3 atm).

Introduction of helium increases the burning rate, at higher helium concentration a third stage quasi-steady burn is observed



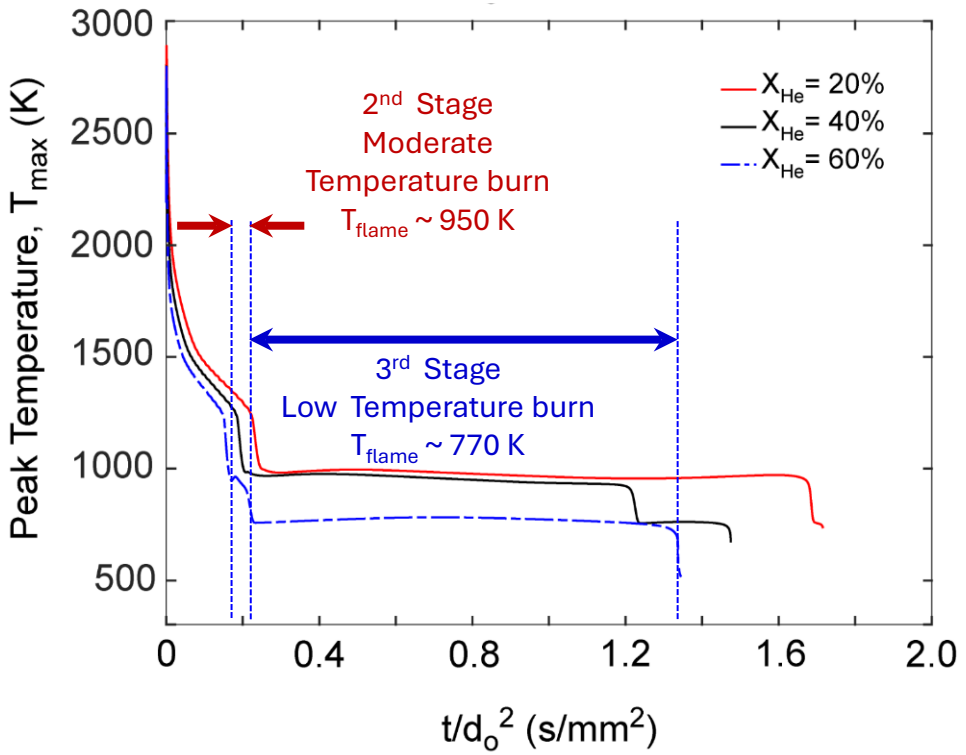
Evolution of peak temperature (n-heptane, d_o = 4.00 mm, X_{O2} = 21%, He exchange with balance N_2, 3 atm).

High Pressure Helium Exchange – 3 atm



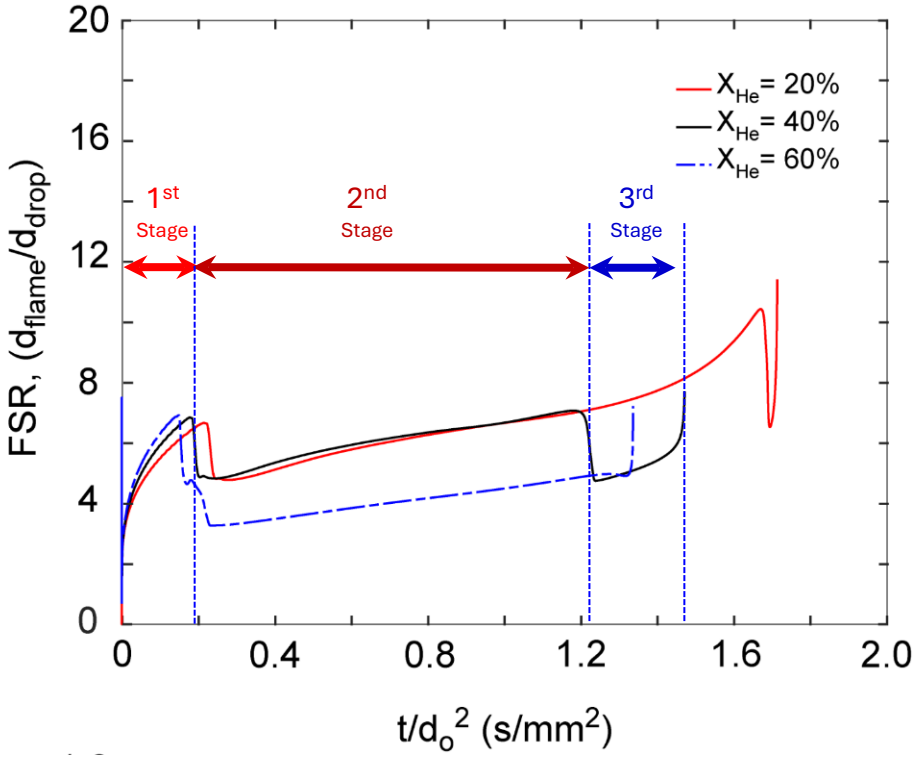
Evolution of droplet diameter (n-heptane, d_o = 4.00 mm, X_{O2} = 21%, He exchange with balance N_2, 3 atm).

- The second stage suppresses at a higher helium concentration
- The droplet diameter regression does not show any significant changes



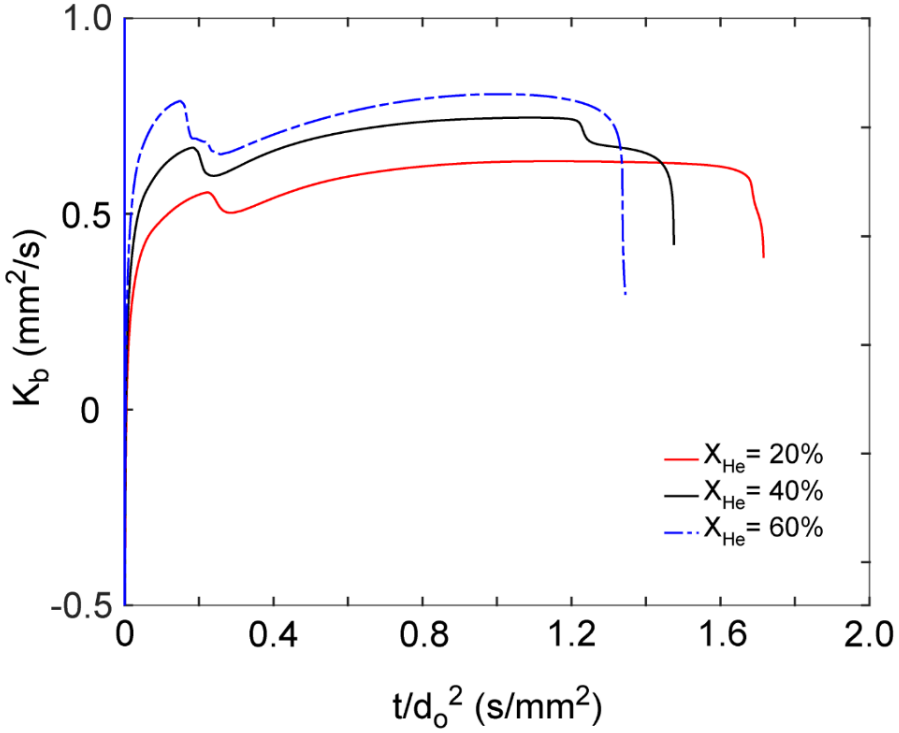
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High Pressure Helium Exchange – 3 atm



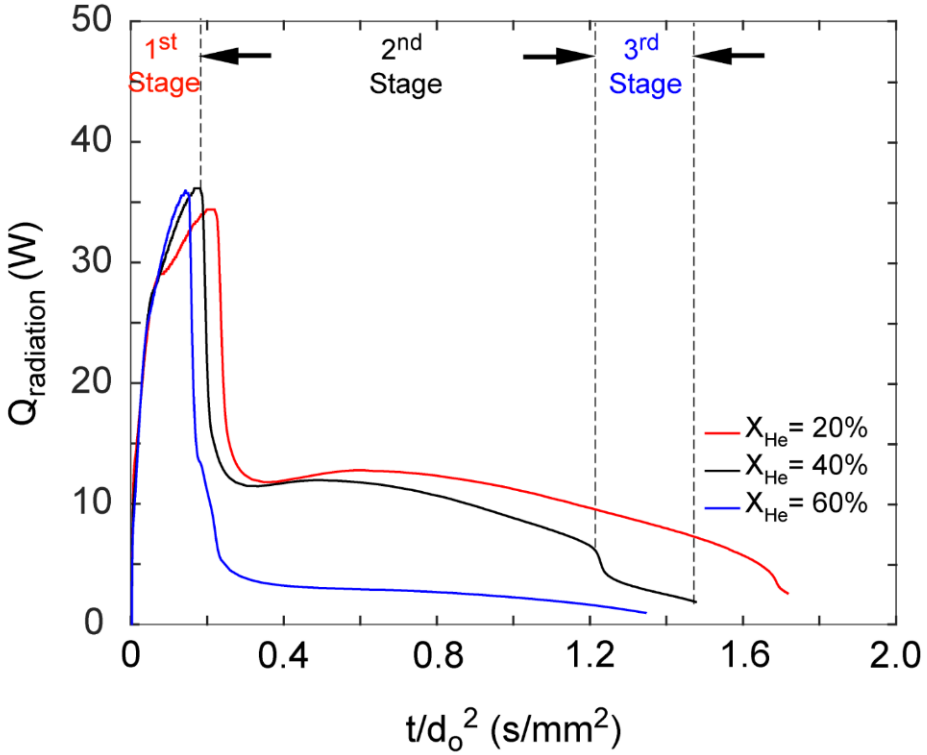
Evolution of flame standoff ratio (n-heptane, d_o = 4.00 mm, X_{O2} = 21%, He exchange with balance N_2, 3 atm).

- A second contraction in the flame occurs as the system transitions to the third stage/low temperature burn.
- Despite the decrease in the maximum temperature, it is the flame contraction that makes the droplet diameter regression insensitive to the different burn stages.



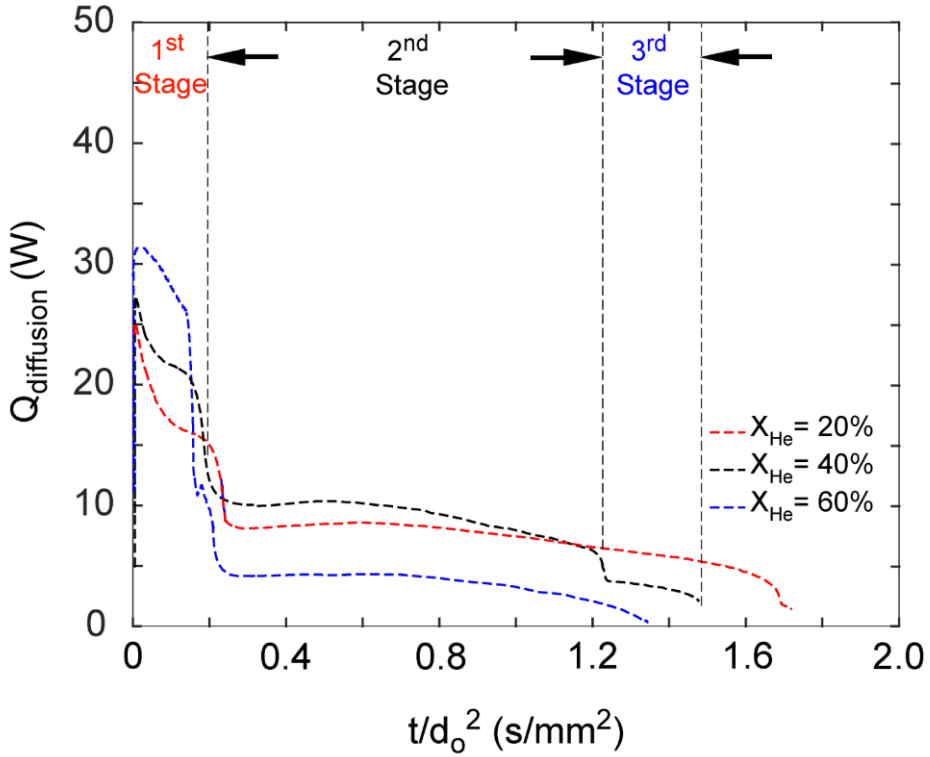
Evolution of burning rate (n-heptane, d_o = 4.00 mm, X_{O2} = 21%, He exchange with balance N_2, 3 atm).

High Pressure Helium Exchange – 3 atm



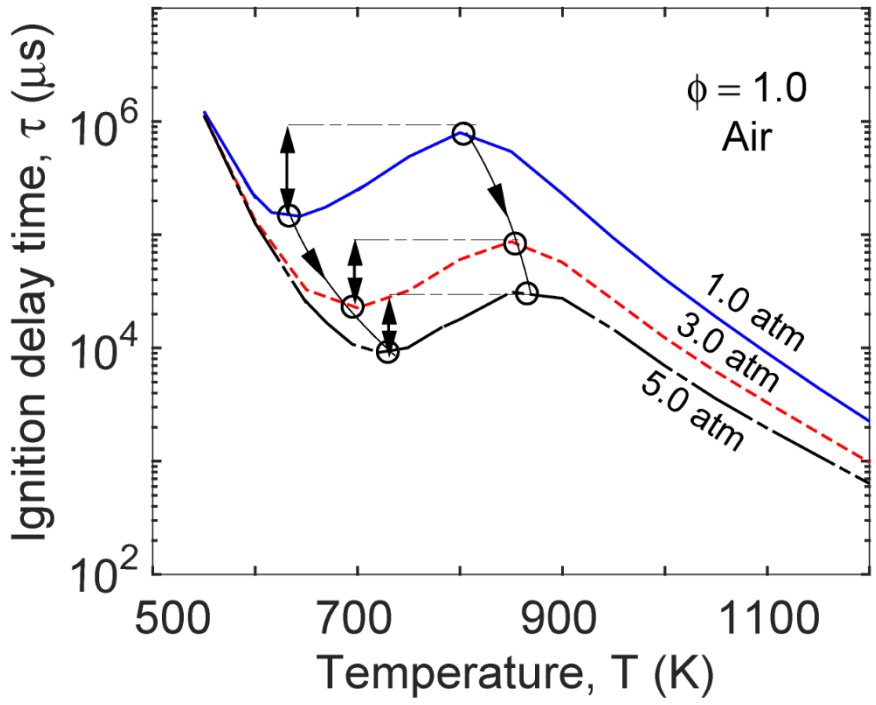
Evolution of radiative loss (n-heptane, d_o = 4.00 mm, X_{O2} = 21%, He exchange with balance N_2, 3 atm).

- A strong coupling between heat generation and losses in the system.
- The radiative and diffusive heat losses are comparable in the first and second stages.
- The diffusive losses become prominent in the third stage, “Cool flame” burn.

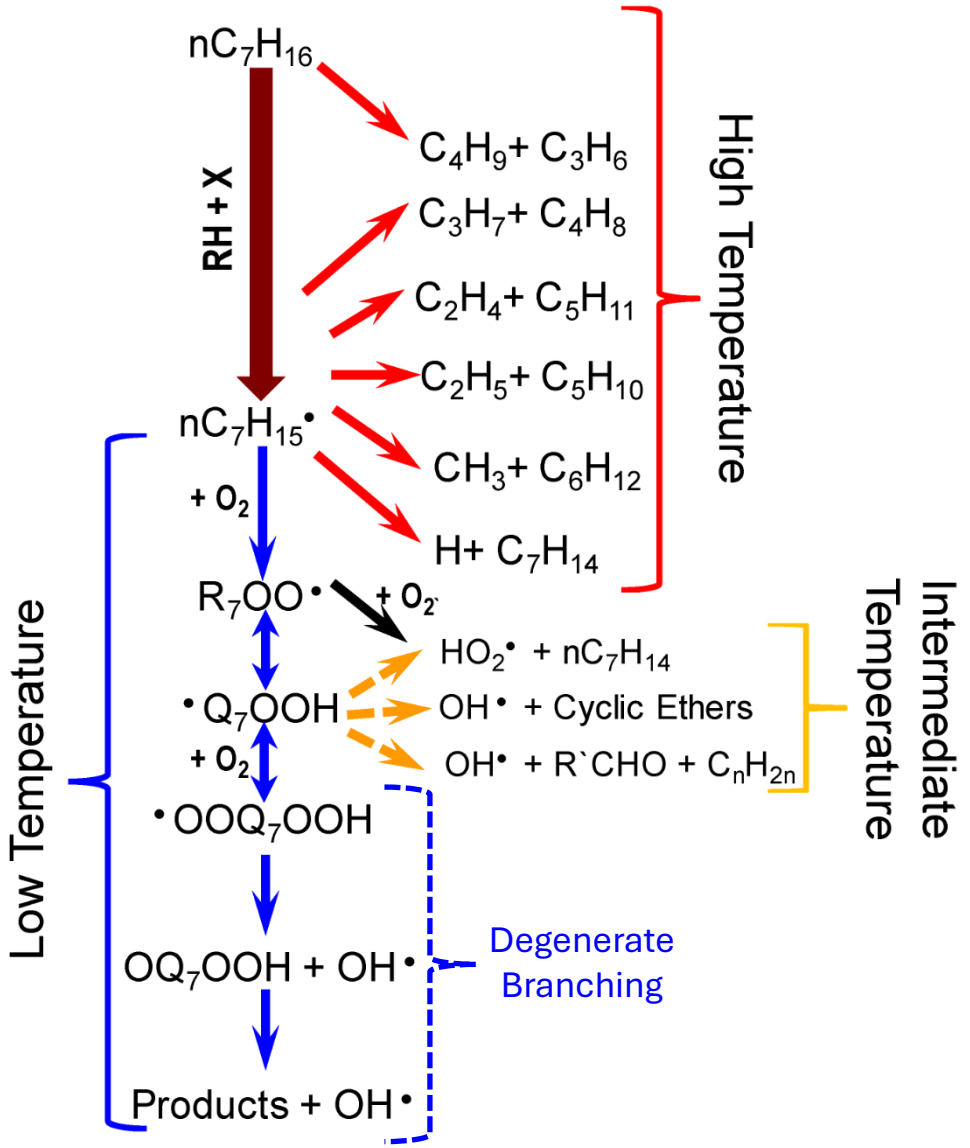


Evolution of diffusive loss (n-heptane, d_o = 4.00 mm, X_{O2} = 21%, He exchange with balance N_2, 3 atm).

Pressure dependence

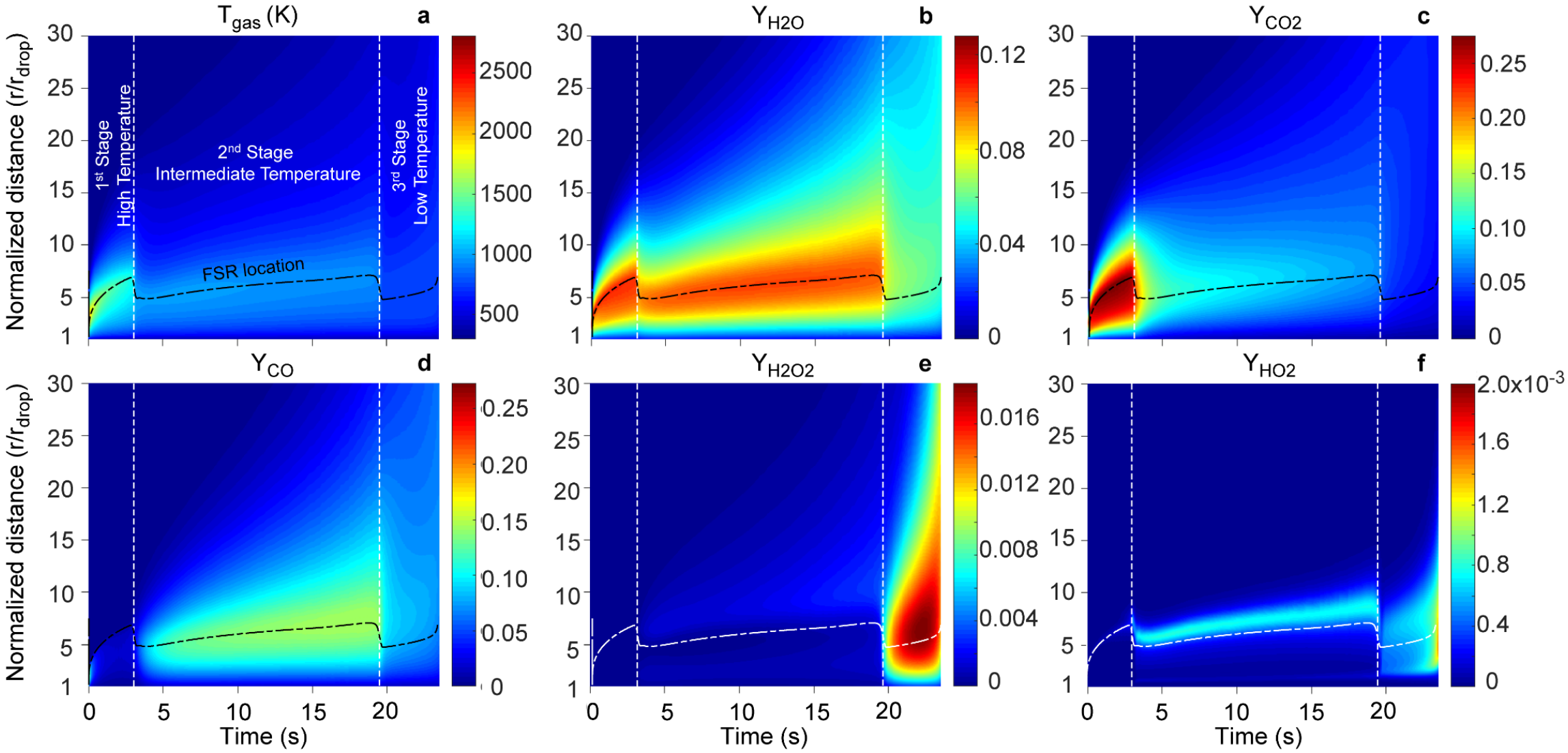


Ignition delay of n-heptane



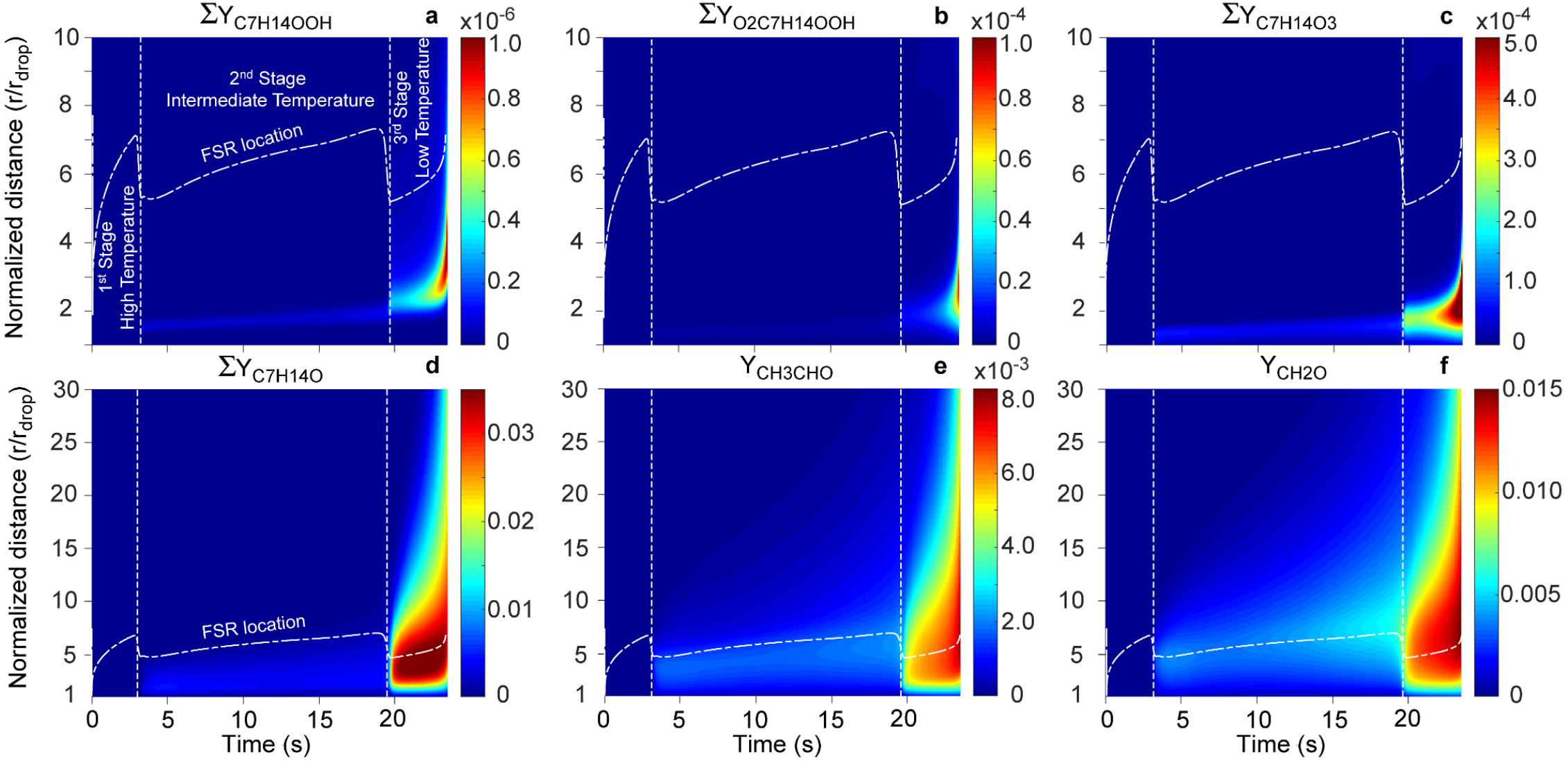
Spatiotemporal variation of species and temperature

Species Distribution – 3 atm, X_{He} = 40%



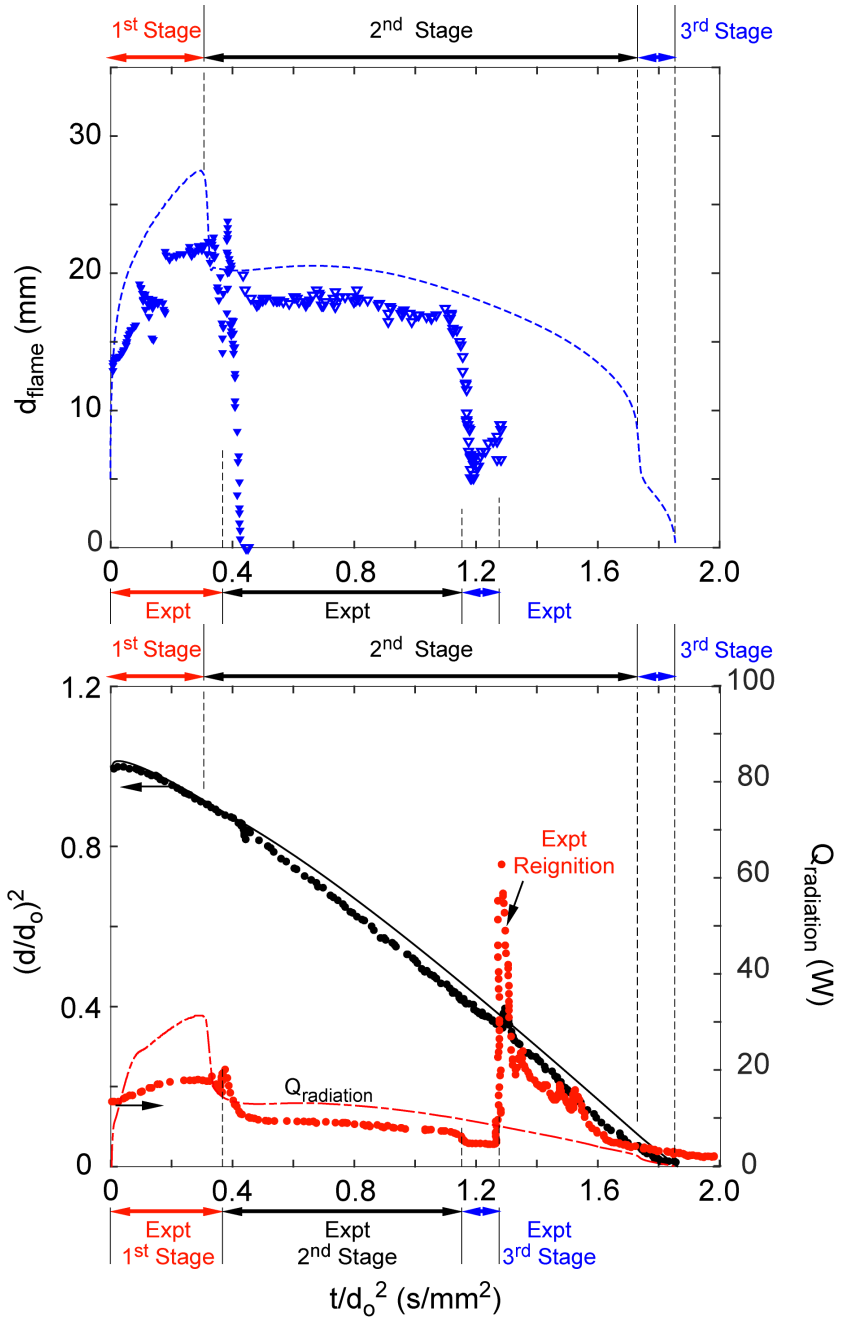
Spatiotemporal distribution of species representative of three distinct stages

Species Distribution – 3 atm, X_{He} = 40%



- Species representative of low temperature kinetics is prominent in the third stage
- Competition between O_2 addition to QOOH and decomposition of QOOH (i.e., OH, HO_2, olefin, carbonyl, QO)

Experimental Observation of Three Stage Burn

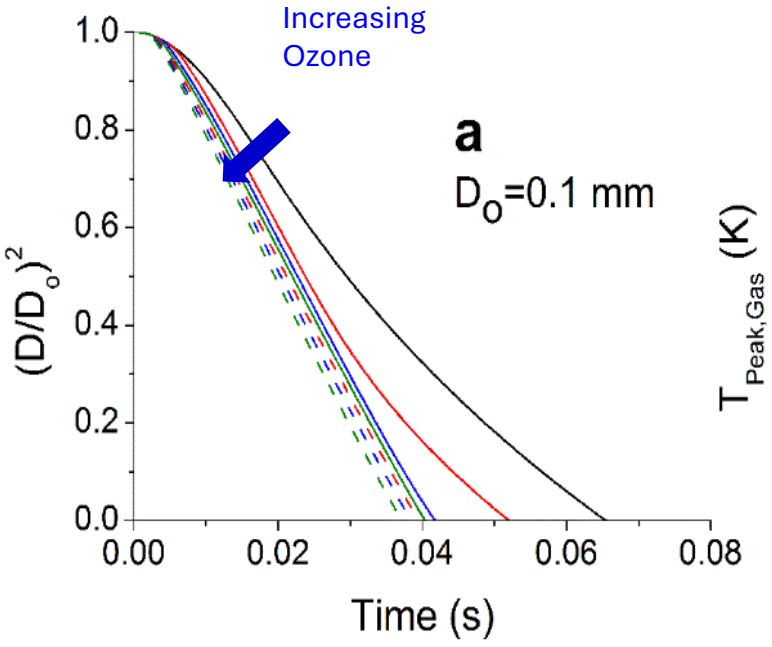


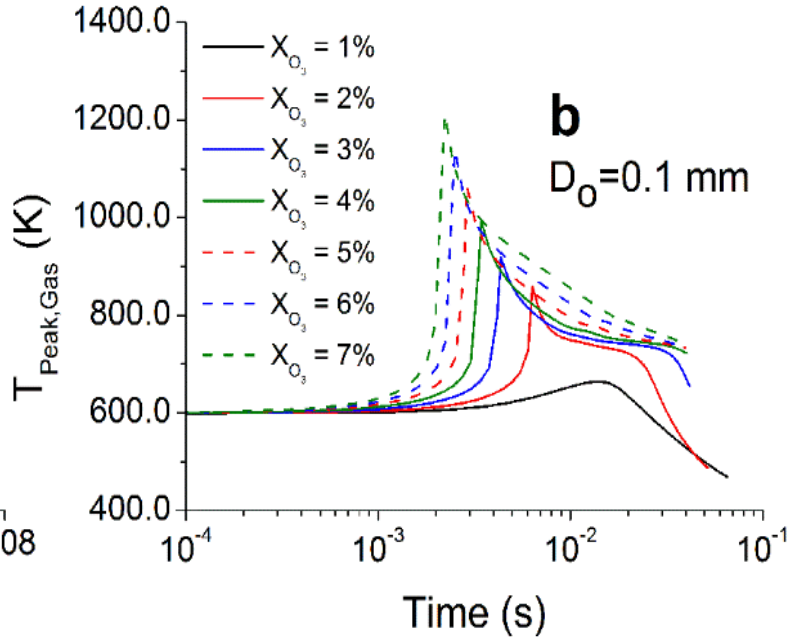
n-dodecane, 2.75 atm, 21.6% O_2, 50.2% N_2, 28.2% He.

Qualitative agreement between predictions and observations

Can We Trigger “Cool Flame” Burn For Smaller Droplets

Ozone Assisted Cool Flames





Amount of Ambient Ozone
➔ Dictating Factor !!!

Low Temperature Combustion



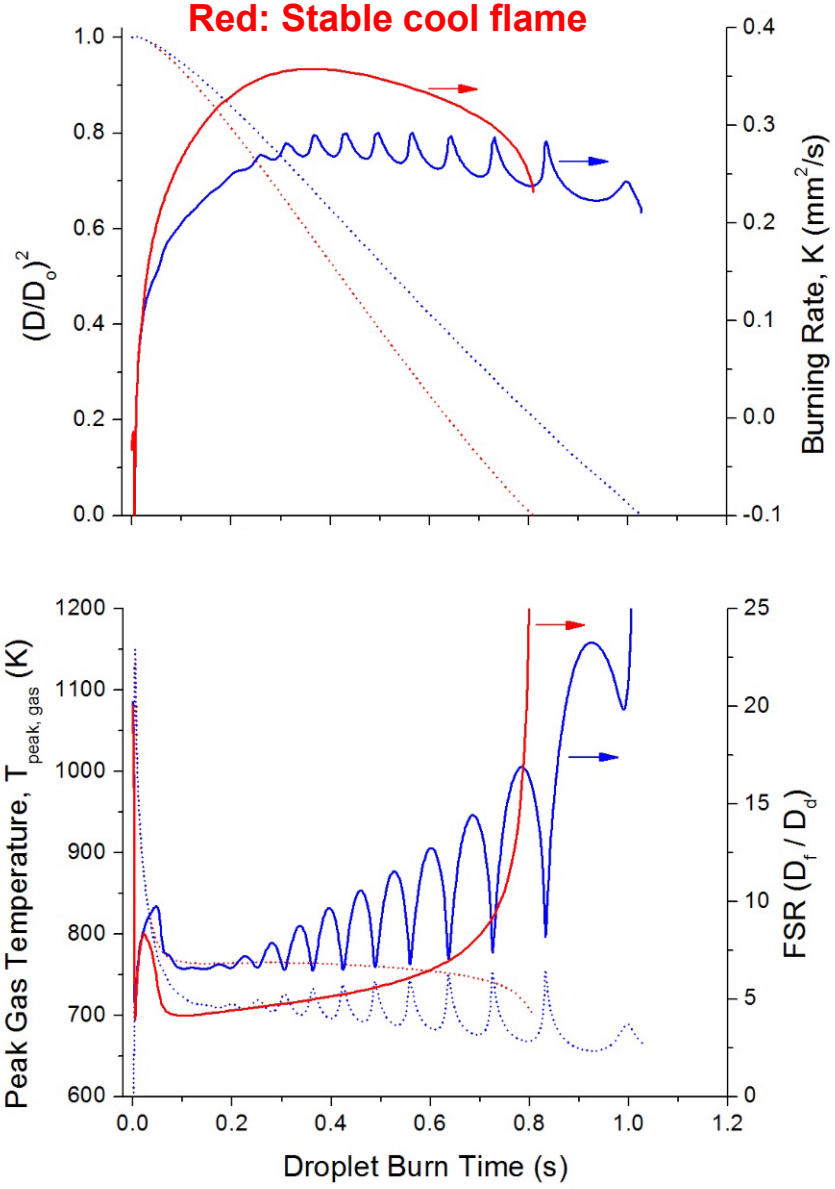
Hot Ignition

Sustained combustion of "Droplet" requires a minimum Ozone Concentration

Reaction
$O_3 + O_2 \rightarrow O_2 + O + O_2$
$O_3 + O \rightarrow O_2 + O + O$
$O_3 + O_3 \rightarrow O_2 + O + O_3$
$O_3 + N_2 \rightarrow O_2 + O + N_2$
$O_2 + O + O_2 \rightarrow O_3 + O_2$
$O_2 + O + N_2 \rightarrow O_3 + N_2$
$O_2 + O + O \rightarrow O_3 + O$
$O_2 + O + O_3 \rightarrow O_3 + O_3$
$O_2 + O_2 \rightarrow O + O + O_2$
$O_2 + O \rightarrow O + O + O$
$O_2 + O_3 \rightarrow O + O + O_3$
$O_2 + H_2O \rightarrow O + O + H_2O$
$O + O + O_2 \rightarrow O_2 + O_2$
$O + O + N_2 \rightarrow O_2 + N_2$
$O + O + O \rightarrow O_2 + O$
$O + O + O_3 \rightarrow O_2 + O_3$
$O_2 + O_2 \rightarrow O_3 + O$
$O_3 + O \rightarrow O_2 + O_2$
$O_3 + H \rightarrow O_2 + OH$
$O_2 + OH \rightarrow H + O_3$
$O_3 + OH \rightarrow HO_2 + O_2$
$O_3 + HO_2 \rightarrow OH + O_2 + O_2$

Cool Flame under Near Limit Conditions

Blue: Oscillatory cool flame
Red: Stable cool flame



X_{O2} = 10%

X_{O3} = 5%

X_{N2} = Balance

P = 1 atm, T = 298 K

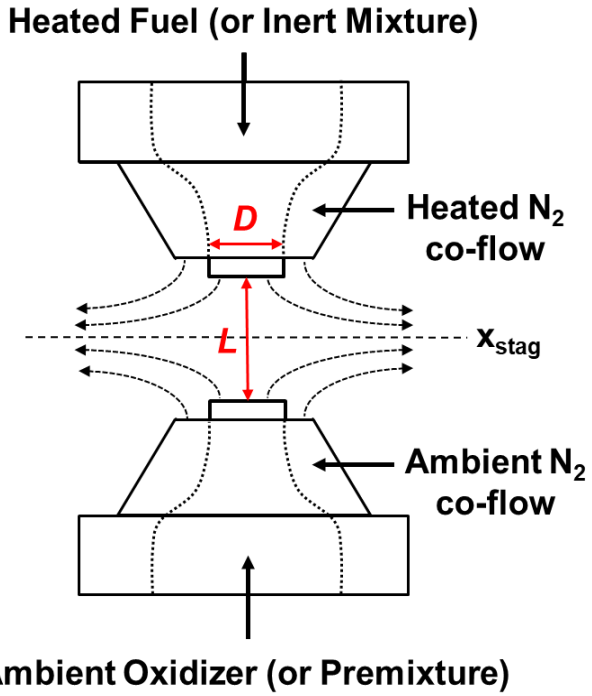
D_o = 0.5 mm

- Flame expands radially ➔ gradual decrease in flame T ➔ Contracts (t < 0.1 s)
- Stable cool flame for brief period of time (~0.12 s @ ~710 K)

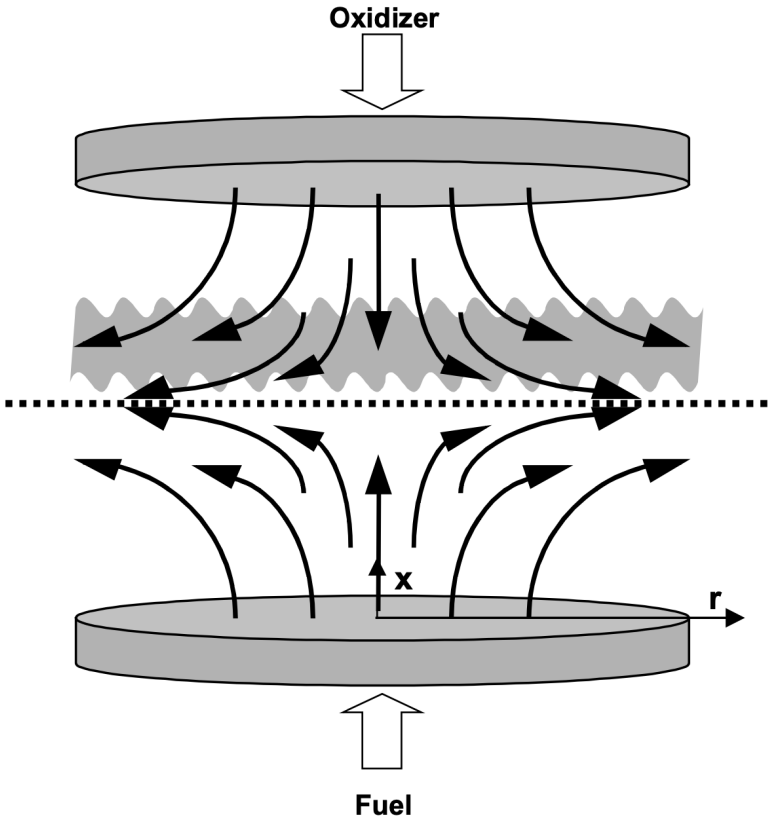
Terrestrial Cool Flame Experiments

Counterflow Diffusion Flame





Cool diffusion flame, **1 g** (Won et al.)



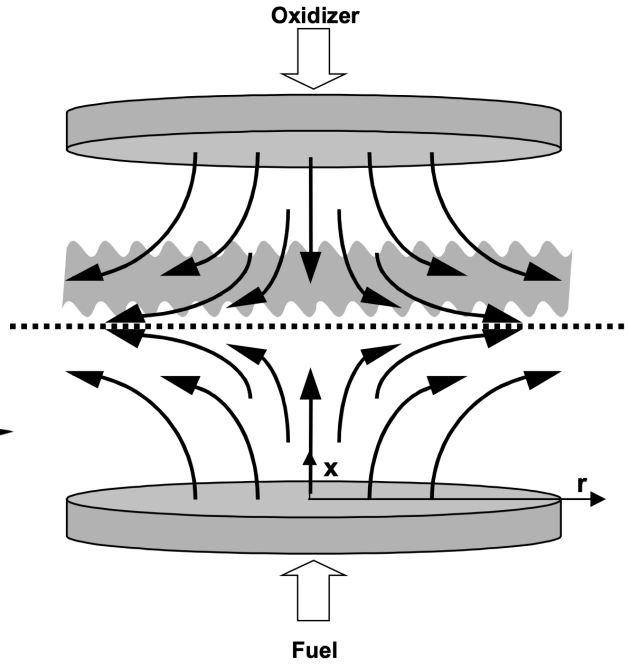
- Excellent configuration for studying flame's strength under the effect of stretch.
- Useful platform for validating reaction kinetics and transport models.
- Idealized condition, it can be treated as a 1D problem:

  ✓ Leads to the derivation of an analytical solution
  ✓ Makes numerical modeling tractable

Limitations of the conventional numerical approach

- The numerical effort to study counterflow flame configuration has been predominantly centered upon OPPDIF module of CHEMKIN chemical kinetic package
- This module implements a similarity transformation approach that reduces the flow field to a one-dimensional/ plug flow problem with the following assumptions:

  ✓ Radial component of velocity is linear in radius
  ✓ Dependent variables become functions of the axial direction only



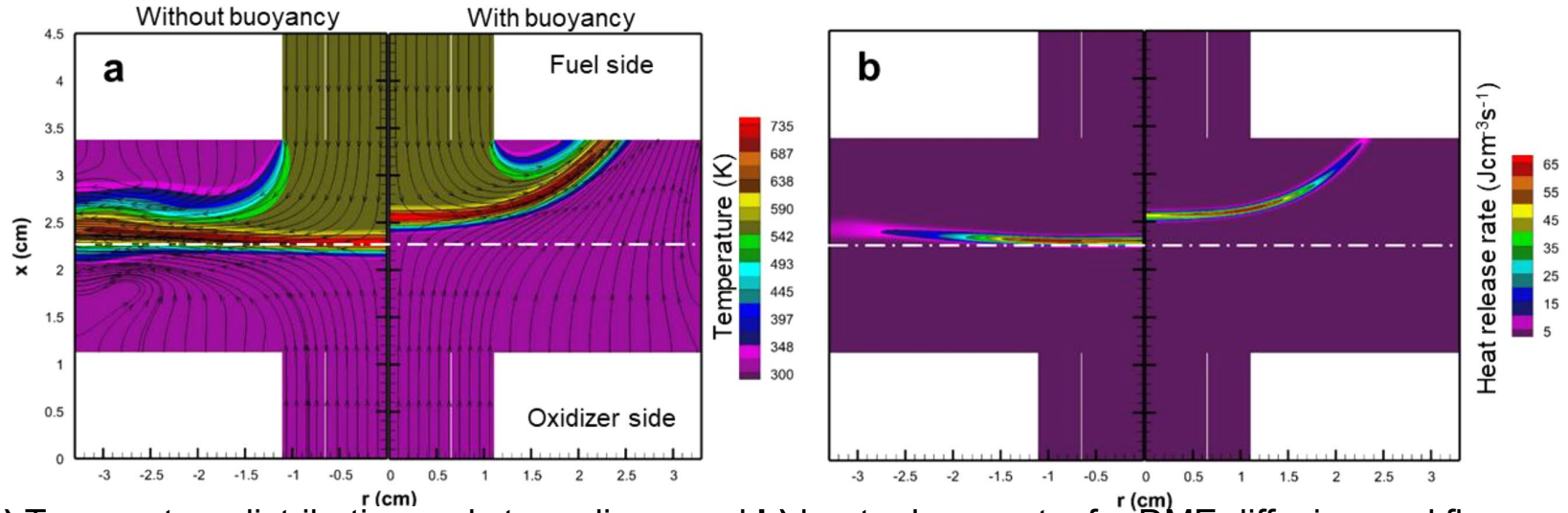
Cool flames

What are the physical conditions that can be represented by this assumption?

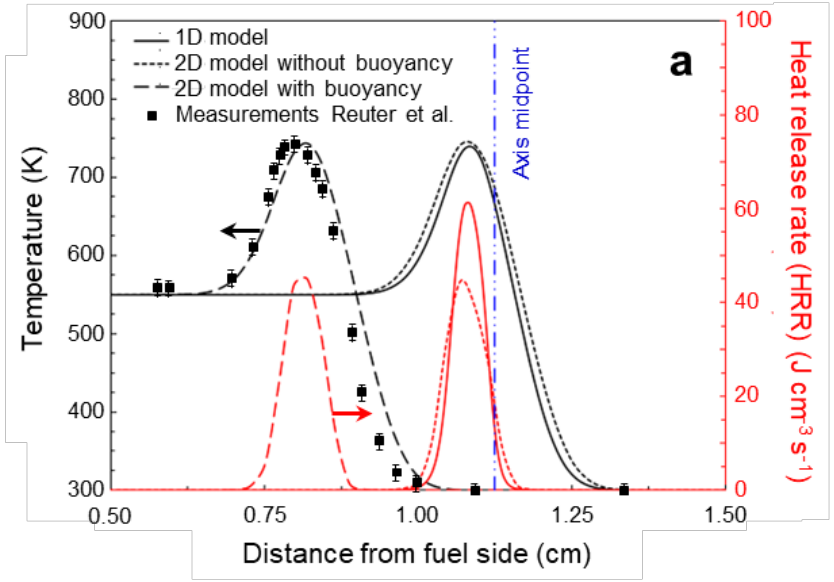
✓ High Reynolds number flow configuration: High strain rate (a)
✓ Thin flame sheet: Hot flame (high activation energy)

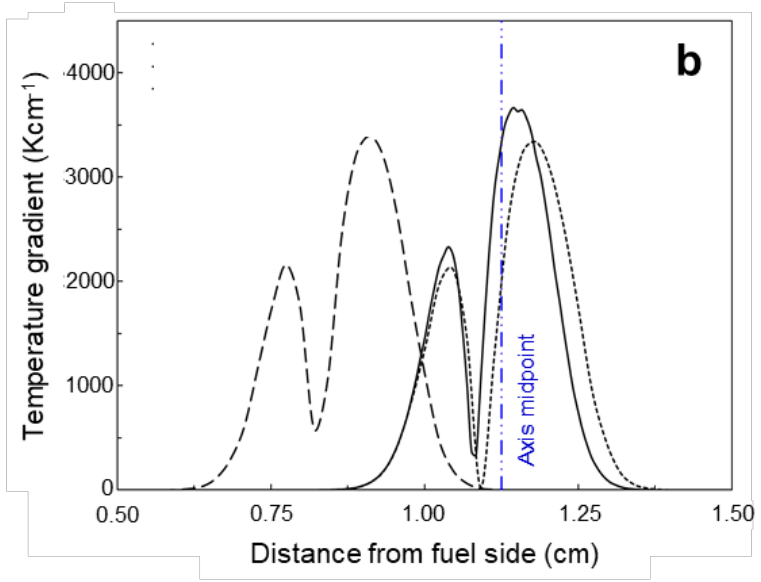
- **Low Re: Low strain rate (a ~100 s^{-1})**
- **Finite Flame Thickness: low activation energy)**

Effect of buoyancy



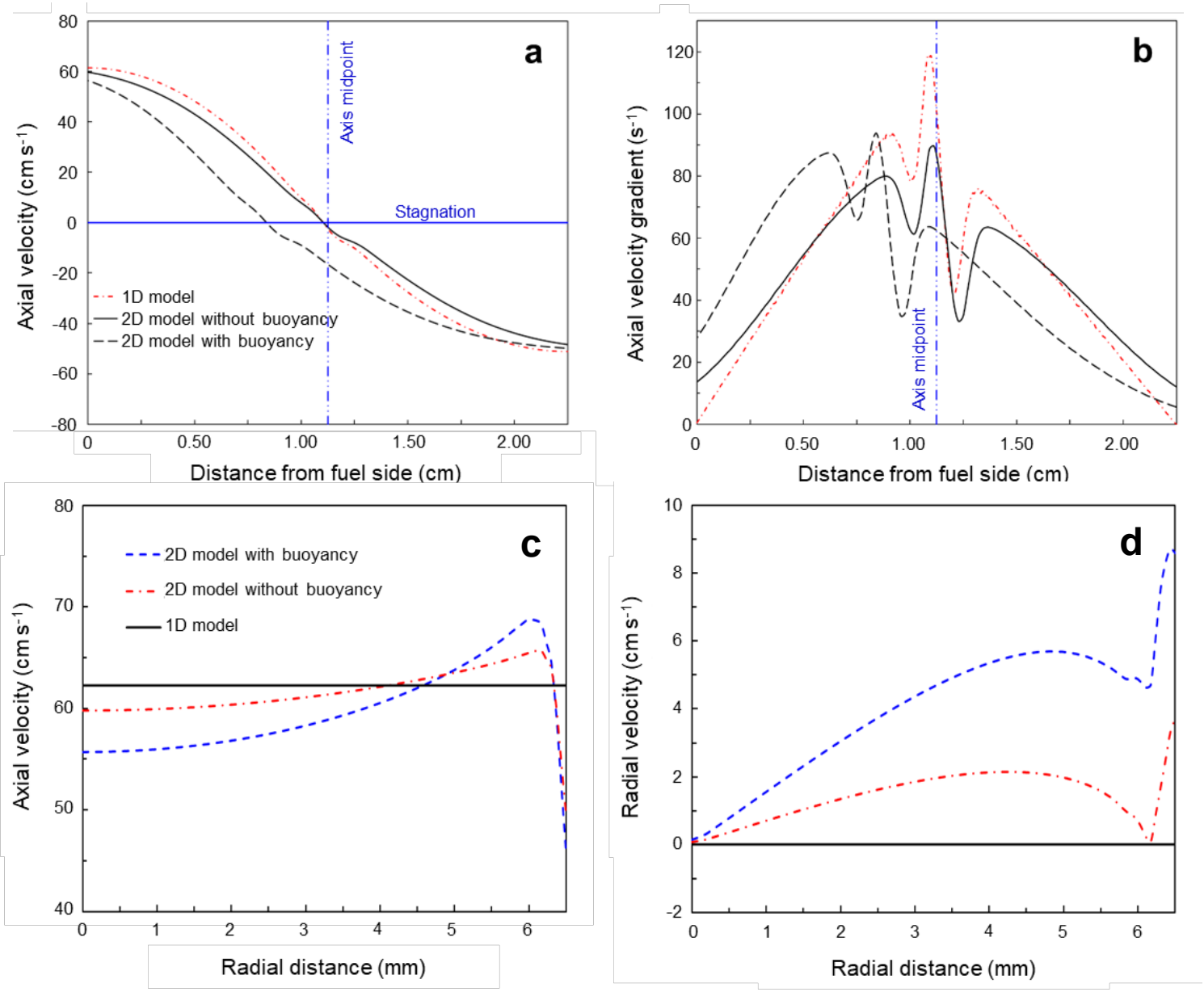
a) Temperature distribution and streamlines, and **b)** heat release rate, for DME diffusion cool flame with and without buoyancy for identical conditions (X_{DME}= 0.587, U_F = 62.3 cm/s, U_O = 50.9 cm/s, ***a*** ~ 90 s-1, P = 1 atm)





- Buoyancy lifts the flame upward.
- Without buoyancy, 1D and 2D models predict the flame at or near the stagnation plane due to momentum-balanced BC.
- The peak flame temperature in 1D is slightly lower compared to the 2D prediction, but the peak HRR shows the opposite trend.
- Temperature gradient is higher in 1D calculation. More diffusive heat loss.

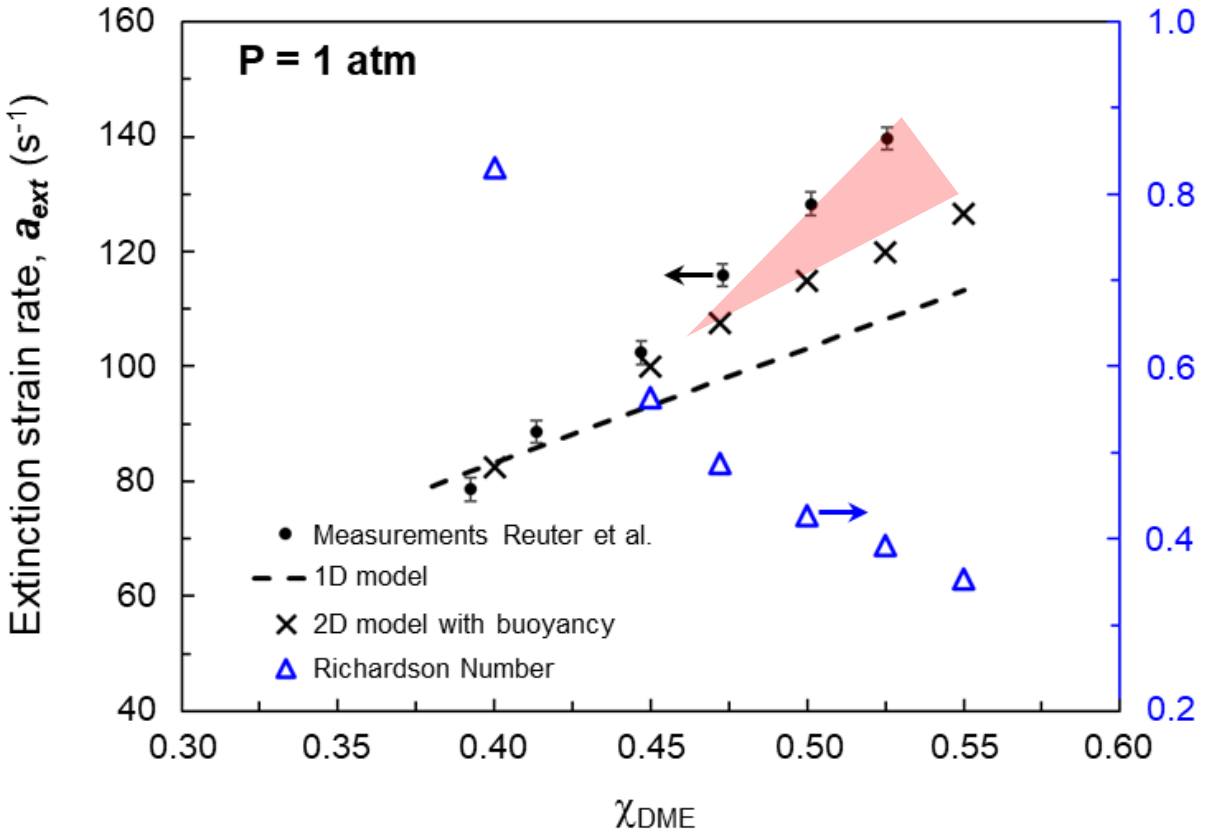
Flow field analyses

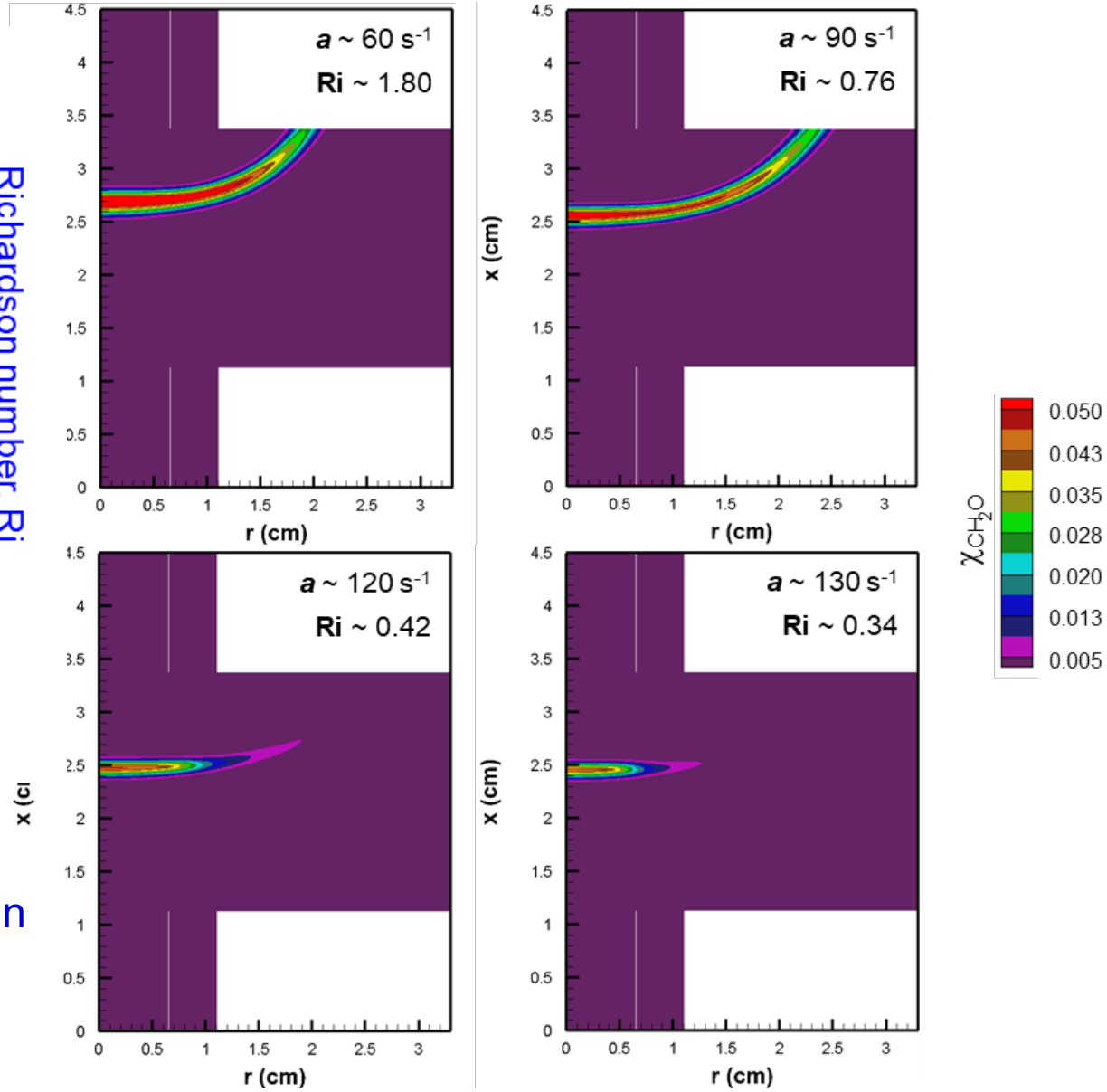


Computed centerline distribution of **a)** axial velocity, **b)** axial velocity gradient
Computed radial distribution of **c)** axial, and **d)** radial velocity at the fuel-side nozzle exit

- The extended entrance channels allow the velocity at the nozzle exits to be perturbed by the flame.
- 2D model captures the perturbation in a self-consistent manner.
- 2D model without buoyancy retains the momentum-balanced condition and forms the stagnation plane at the axis midpoint, exhibiting identical perturbation on either side
- Axisymmetric dip in 2D simulation, observed in experiments of Sarnecki et al.

Effect of global strain rate





- Flame is pushed away from the nozzle as the convective effect becomes dominant with the increase of global strain rate.
- The flame location ceases to move further when the buoyancy effect becomes negligible (i.e., low Ri).

X_{CH2O} contours of DME diffusion cool flame at different global strain rates (X_{DME} = 0.587, P = 1 atm)

Summary

- Insights obtained from modeling on cool flame characteristics and dynamics
- Models were developed for a number of scenarios
- Models were validated against the FLEX data and then utilized to explore other scenarios
- Extensive amount of experimental data available for interpretation and new insights

The Resurgence of “Cool Flame” Research

Combustion and Flame 159 (2012) 3583–3588

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Can cool flames support quasi-steady alkane droplet burning?

Vedha Nayagam [a,*], Daniel L. Dietrich [b], Paul V. Ferkul [a], Michael C. Hicks [b], Forman A. Williams [c]

[a] National Center for Space Exploration Research, Cleveland, OH 44135, USA
[b] NASA Glenn Research Center, Cleveland, OH 44135, USA
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ABSTRACT

Experimental observations of anomalous combustion of n-heptane droplets burning in microgravity are reported. Following ignition, a relatively large n-heptane droplet first undergoes radiative extinction, that is, the visible flame ceases to exist because of radiant energy loss. But the droplet continues to experience vigorous vaporization for an extended period according to a quasi-steady droplet-burning law, ending in a secondary extinction at a finite droplet diameter, after which a vapor cloud rapidly appears surrounding the droplet. We hypothesize that the second-stage vaporization is sustained by low-temperature, soot-free, “cool-flame” chemical heat release. Measured droplet burning rates and extinction diameters are used to extract an effective heat release, overall activation energy, and pre-exponential factor for this low-temperature chemistry, and the values of the resulting parameters are found to be closer to those of “cool-flame” overall reaction-rate parameters, found in the literature, than to corresponding hot-flame parameters.

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1. Introduction

For over 50 years, scientists have been studying droplet combustion in microgravity, beginning with the modest one-second drop-tower experiments by Kumagai [1] in Japan in 1957. For heptane droplets, his measurements verified attainment of quasi-steady combustion, in which the square of the droplet diameter decreases linearly with time, in accord with theory. More recently, with the availability of long-duration space-based microgravity platforms, researchers were able to study larger droplets with longer burning histories [2–4]. In these and other experiments, combustion of n-heptane has been extensively investigated, recently in part because of its consideration as a component of surrogate fuels for diesel and gasoline engines [5]. The conventional wisdom thus far has been that small heptane droplets, once ignited, burn to completion, extinguish in a flash as a consequence of impurity or product accumulation, or extinguish diffusively [6] at a finite droplet diameter, whereas larger droplets extinguish radiatively as a result of excessive radiant heat loss from the flame zone [7]. Diffusive extinction of small droplets occurs as a consequence of the diffusion time, which decreases with decreasing droplet diameter, becoming less than the chemical time required to complete the heat release in the gas phase. On the other hand, radiative extinction occurs for large droplets because, as the droplet diameter is increased, the rate of radiant loss, which is proportional to the flame volume, increases with respect to the rate of heat release, which is proportional to the flame diameter. Following either one of these last two modes of extinction, the droplet simply evaporates in the hot environment, without any chemical heat release in the gas phase. Theoretical models and numerical simulations [8], with detailed chemical kinetics geared towards high-temperature chemistry, have supported this understanding.

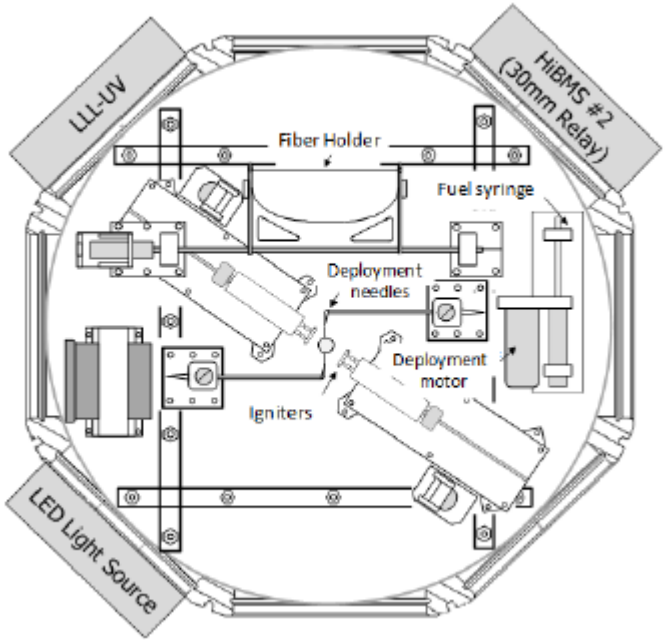
In this article, we report, for the first time, experimental observations in which large n-heptane droplets exhibit dual modes of combustion and extinction. First the droplet undergoes normal burning with a visible flame surrounding the droplet and extinguishes radiatively at a relatively large droplet size. This event is followed by continued rapid quasi-steady vaporization of the droplet without any visible flame detected, which ends abruptly, at a point called “second-stage extinction” here, leaving behind a smaller droplet, which then either experiences normal time-dependent evaporation in the hot surrounding environment or sometimes apparently grows slightly through condensation of the vapor in the cloud that forms upon extinction, or through migration of the vapor-cloud particles to the droplet surface.

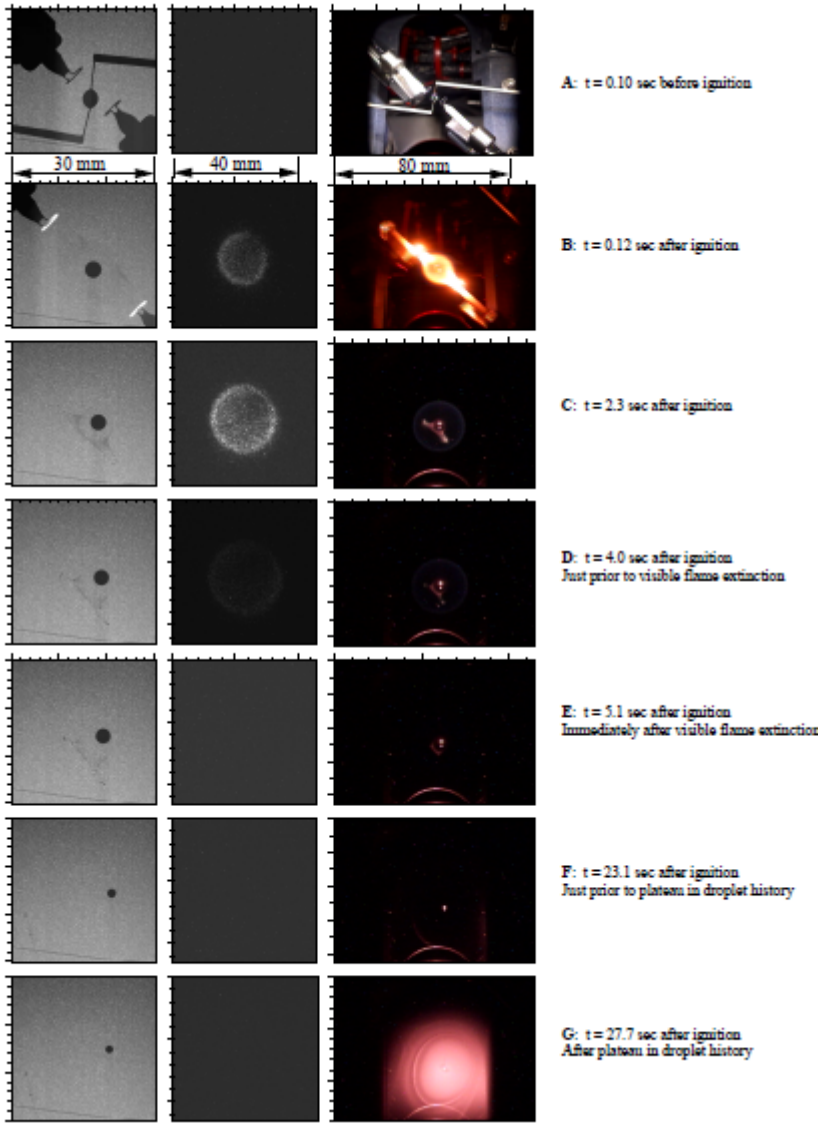
Two-stage burning has been reported and studied previously for bi-component fuel droplets (e.g., [4]), explained qualitatively by the lighter component supporting the initial flame, with the second period involving combustion of the heavier component, and both stages involve traditional hot flames. Dual-mode combustion has not previously been discussed for pure-fuel droplets, nor have

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“Anomalous Combustion/Cool Flame?” of large n-heptane droplet onboard ISS in 2011, NASA **FLEX**





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Isolated *n*-heptane droplet combustion in microgravity: “*Cool Flames*” – Two-stage combustion

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ABSTRACT

Recent experimentally observed two stage combustion of *n*-heptane droplets in microgravity is numerically studied. The simulations are conducted with detailed chemistry and transport in order to obtain insight into the features controlling the low temperature second stage burn. Predictions show that the second stage combustion occurs as a result of chemical kinetics associated with classical premixed “*Cool Flame*” phenomena. In contrast to the kinetic interactions responsible for premixed cool flame properties, those important to cool flame droplet burning are characteristically associated with the temperature range between the turnover temperature and the hot ignition. Initiation of and continuing second stage combustion involves a dynamic balance of heat generation from diffusively controlled chemical reaction and heat loss from radiation and diffusion. Within the noted temperature range, increasing reaction temperature leads to decreased chemical reaction rate and vice versa. As a result, changes of heat loss rate are dynamically balanced by heat release from chemical reaction rate as the droplet continues to burn and regress in size. At reaction temperatures below the turnover, heat loss over takes the heat release rate and extinction occurs. Should heat release exceed heat loss as the temperature increases to that for hot ignition, initiation of a high temperature burning phase may be possible. Parametric study on factors leading to initiation of the second stage burning phenomena are studied. Results show that both carbon dioxide and helium diluents can promote initiation of low temperature burning at smaller initial drop diameters than found with nitrogen as diluent. Small amounts of carbon dioxide and helium in the ambient is sufficient to activate the phenomena. The chemical kinetics dictating the second stage combustion and extinction process is also discussed.

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1. Introduction

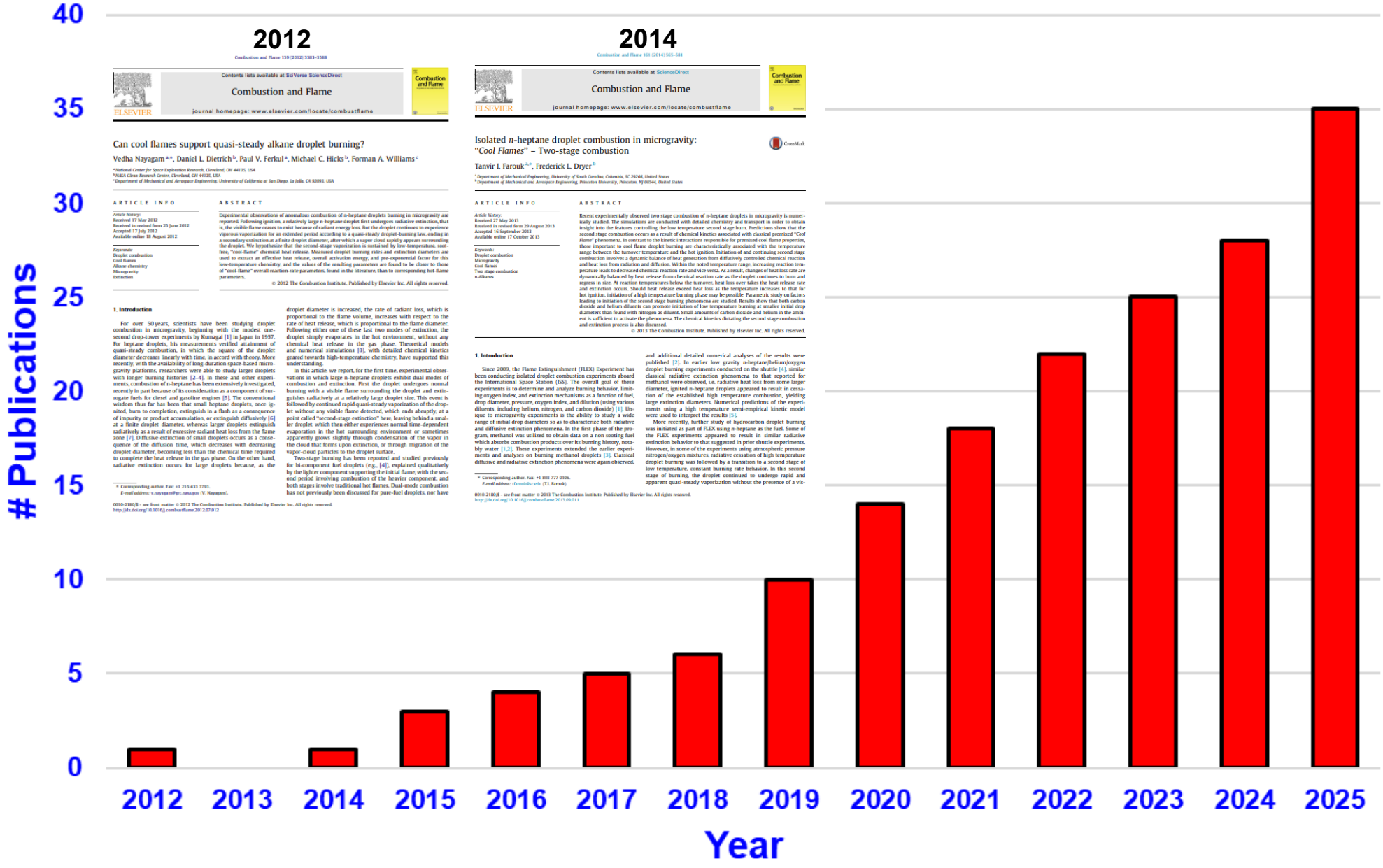
Since 2009, the Flame Extinguishment (FLEX) Experiment has been conducting isolated droplet combustion experiments aboard the International Space Station (ISS). The overall goal of these experiments is to determine and analyze burning behavior, limiting oxygen index, and extinction mechanisms as a function of fuel, drop diameter, pressure, oxygen index, and dilution (using various diluents, including helium, nitrogen, and carbon dioxide) [1]. Unique to microgravity experiments is the ability to study a wide range of initial drop diameters so as to characterize both radiative and diffusive extinction phenomena. In the first phase of the program, methanol was utilized to obtain data on a non sooting fuel which absorbs combustion products over its burning history, notably water [1,2]. These experiments extended the earlier experiments and analyses on burning methanol droplets [3]. Classical diffusive and radiative extinction phenomena were again observed, and additional detailed numerical analyses of the results were published [2]. In earlier low gravity *n*-heptane/helium/oxygen droplet burning experiments conducted on the shuttle [4], similar classical radiative extinction phenomena to that reported for methanol were observed, i.e. radiative heat loss from some larger diameter, ignited *n*-heptane droplets appeared to result in cessation of the established high temperature combustion, yielding large extinction diameters. Numerical predictions of the experiments using a high temperature semi-empirical kinetic model were used to interpret the results [5].

More recently, further study of hydrocarbon droplet burning was initiated as part of FLEX using *n*-heptane as the fuel. Some of the FLEX experiments appeared to result in similar radiative extinction behavior to that suggested in prior shuttle experiments. However, in some of the experiments using atmospheric pressure nitrogen/oxygen mixtures, radiative cessation of high temperature droplet burning was followed by a transition to a second stage of low temperature, constant burning rate behavior. In this second stage of burning, the droplet continued to undergo rapid and apparent quasi-steady vaporization without the presence of a vis-

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Resurgence of “Cool Flame” Research



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Thank You